The value of being indispensable: a cooperative approach to bargaining with a continuum of players*

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Abstract

This paper investigates the idea of using random order values, a concept from cooperative game theory, to model bargaining outcomes in games with one indispensable player and a continuum of fringe players. Two important special cases of random order values are discussed in detail: the Shapley value and the weighted value. I show that this approach proves to be very tractable, and leads to results that are in line with one's intuitive expectations about bargaining outcomes. Thus, it can be a useful tool in modelling situations where one does not want to attribute all the bargaining power to one side of the market, and at the same time would like to avoid having to deal with an explicit multi-stage bargaining game. The utility of this approach is demonstrated via a simple model of a two-sided market.

1 Introduction

Using cooperative game theory to model bargaining outcomes has quite a few precedents in the industrial organization literature. This is in part motivated by the appealing properties of the resulting gain distributions and their tractability. Furthermore, this practice also has solid theoretical justifications thanks to the long line of studies showing that various cooperative solution concepts are related to certain dynamic, non-cooperative bargaining games. For example, Gul (1989), Winter (1994), S. Hart and Mas-Colell (1996), and Inderst and Wey (2003) or, perhaps most relevant to this paper, Stole and Zwiebel (1996a) all propose various extensive-form bargaining games in which the equilibrium corresponds to players' Shapley-values.

For practical applications, the case of one indispensable player and a large number of small ones is particularly important in a number of settings. Examples include intra-firm ownership structures (O. Hart and Moore 1990), wage bargaining (Stole and Zwiebel 1996a; Stole and Zwiebel 1996b; Levy and Shapley 1997), collusion and mergers (Segal 2003) and upstream-downstream supply chains (Inderst and Wey 2003; Montez 2007). The study of multi-sided markets (or platforms) is also a good fit for this modeling approach, as the platform is often assumed to be indispensable for the various sides to interact.[1]

A common feature of most of the aforementioned papers is that they assume a finite number of players. This paper explores the idea of using an infinite-player version of these models, where fringe players are represented as a continuum. As is often the case in economics, this infinite-player approximation

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[1] Huang and Xie (2022) is amongst the very few examples of using cooperative game theory in this setting.

turns out to be more tractable than the original, finite-player model, while retaining the main features of the latter. For example, the resulting profit shares depend on the production function similarly to the finite-player case, while becoming continuous and differentiable functions of the size of the fringe. This can be useful when embedding these results into more complex models.

The main contribution of this paper is to provide a general framework for modeling bargaining outcomes in games with one indispensable player and a continuum of fringe players. I propose an oceanic game with the main feature that the large player is indispensable for creating value[2], and examine the (Shapley, weighted and random order) values of the players in this game. I demonstrate that these values are described by very tractable expressions and have illuminating geometric interpretations. Just as importantly, I show that the outcomes are in line with one's intuitive expectations of bargaining power in such situations.

On one hand, this paper provides a taxonomy of the various results found in some applied papers. Furthermore, it introduces some new results pertaining to the case of multiple types of fringe players. In particular, I highlight how the value of the fringe players can be expressed in terms of their marginal contributions, and how players' bargaining weights affect outcomes through changing the probability weights of orderings from which those marginal contributions are calculated. Finally, to showcase the utility of this approach, I also build a simple model of two-sided markets. My hope is that the results in this paper can be used as a tool-kit for embedding bargaining outcomes into more complex models.

The present paper is closely related to two bodies of literature: a theoretical and an applied one. Cooperative games with a finite number of atomic players and a continuum of non-atomic ones are called mixed or oceanic games (Milnor and Shapley 1978), and have been studied in the cooperative game theory literature. Most papers in this line of research focus on rather fundamental questions, such as for which games does the Shapley-value exist, when is it unique (S. Hart 1973), and whether the various ways of defining it[3] lead to the same result (Fogelman and Quinzii 1980).[4] My aim with this paper is both somewhat more general and much more specific in certain aspects. For one, I focus on the asymptotic approach, and within that, a specific way of approximating the infinite game of interest. This is motivated by the fact that I am not interested in the oceanic game per se, but rather in its ability to approximate finite-player games. Furthermore, I restrict my attention to a specific, but economically important subset of games: those with one indispensable player, and a continuum of fringe players consisting of a finite number of types. These two restrictions in turn allow me to go beyond the Shapley value, and obtain results for a more general set of solution concepts: random order values. Additionally, instead of looking at general results, such as existence, I can provide and interpret explicit expressions for the values of the various players.

The other strand of research that is closely related to this paper belongs to the industrial organization literature. Despite the theoretical appeal and tractability of oceanic games, they, in general, saw remarkably little use in the industrial organization, and, more generally, applied economics. Nevertheless, there are a few prominent exceptions, such as Stole and Zwiebel (1996a) and Stole and Zwiebel (1996b) and Levy and Shapley (1997), which use the Shapley value to model wage bargaining under the assumption of divisible labor (continuum of workers). The first two of these paper also provides microfoundations for using the Shapley value and the weighted value in an intra-firm bargaining setting. The present paper contributes to this literature by generalizing some of its results, and an examination of some general properties of the various values in the context of such games. In particular, in addition to the aforementioned papers, the proposed model includes two new features: (1) results about the distribution

[2]This assumption will be somewhat relaxed throughout some of the generalizations.

[3]One can characterize the value of the oceanic game directly, relying on a set of axioms (axiomatic approach). Alternatively the value of an infinite-player game can also be defined as the limit of the value of a series of finite-player games (asymptotic approach).

[4]S. Hart (1973) shows that in the case of mixed games, the usual axioms do not in general characterize a unique value. However, Fogelman and Quinzii (1980) demonstrates that, for a quite large subset of games, the asymptotic approach leads to one specific element from the set characterized by the axiomatic approach.

of the value between the various types of fringe players, and (2) using random order values instead of the less general Shapley value or weighted value. Furthermore, I also provide results for the intersection of two important extensions in Stole and Zwiebel (1996a): the case of weighted values in the case of multiple types of fringe players.

Finally, en example application is presented to demonstrate the usefulness of the proposed approach. This model is based on the seminal work of Armstrong (2006) on multi-sided markets. Thus, although the present paper is mainly theoretical, it also has some minor contributions towards better understanding the functioning of multi-sided markets.

The paper is organized as follows. In Section 2, I present the main results for the case of one indispensable player and a continuum of identical fringe players. Section 3 extends these results to the case of multiple types of fringe players. This is where the most novel results of the paper are presented. Section 4 provides an example application of the results to a simple model of a two-sided market. Finally, Section 5 concludes the paper.

2 Identical fringe players

In this section I examine the case of identical fringe players. Many of these results are known in the literature, but I include them here for three reasons. First, to provide build intuition for the more general case later on. Second, to highlight the similarities to the multi-sided case, and to provide a bridge between the two. Finally, I also express the fringe's value in terms of its weighted marginal contributions, which is a useful result in its own right. In the main text, I only consider the most straightforward case (one large completely indispensable player), but in 2.3 I will somewhat relax these assumptions.

Let us start by defining the cooperative game that describes this situation. Consider a game with two types of players: a major player P and n smaller players A_i, $1 \leq i \leq n$. Formally, let us denote the set of all players as $N = \{P, A_1, \ldots, A_n\}$. Assume that (1) no coalition of players can achieve a positive value without the participation of P and (2) players A_i are identical to each other. Let $n_T(S)$ denote the number of players of type $T \in \{P, A\}$ in coalition S. Under these assumptions, the characteristic function of this game (the value that a coalition $S \subset N$ can achieve) is the following:

$$v(S) = \begin{cases} 0 & \text{if } P \notin S \\ f\left(\frac{n_A(S)}{n}\right) & \text{otherwise.} \end{cases}$$

Such a game can represent, for example, a (one-sided) platform and a number of potential entrants, where the entrants can only reach their prospective consumers through the platform, or as a vertical supply chain with one upstream producer and a number of downstream sellers.

Before delving into the various values, let us characterize the conditions under which this game is monotone and superadditive. Monotonicity requires that the value of any coalition is weakly higher than the value of any of its subsets. Superadditivity on the other hand requires that the value of the union of two disjoint coalitions is weakly higher than the sum of their values. The latter is a key property of cooperative games, as it ensures that it always pays off to form the grand coalition (i.e., the coalition that includes all players). Furthermore, in the context of bargaining, these properties provide some theoretical support for using the Shapley-value or the weighted value to represent bargaining outcomes, as many results creating a link between non-cooperative bargaining games and the values of cooperative games often rely on these assumptions.[5] In this specific setting, characterizing monotonicity and superadditivity is extremely straightforward.

[5] For example, the results in Gul (1989) rely on superadditivity, while monotonicity suffices for S. Hart and Mas-Colell (1996).

Proposition 1. *The game (N, v) is monotone and superadditive if and only if f is increasing and $f(0) \geq 0$.*

The intuition behind this result is quite straightforward. As player P is necessary to create any value, no two disjoint coalitions can both achieve a positive value. Therefore, superadditivity is equivalent to monotonicity in this case.

In the following, let us assume that the assumptions required for monotonicity are always satisfied (i.e., f is monotone increasing).[6] As a side note, it also ensures that the game is of bounded variation, and thus the main theorem in Fogelman and Quinzii (1980) applies. Namely, the asymptotic and the axiomatic approach of characterizing the Shapley-value of the game both lead to the same result[7]. This result could also be used to prove Proposition 3 instead of the direct proof provided in Appendix C.

2.1 Random order values

Random order values are a class of cooperative solution concepts characterized by efficiency, monotonicity, linearity and the null-player axiom (Weber 1988). That is, they are a generalization of the Shapley value, where the axiom of symmetry is relaxed. Another way to think about them is that they are the average marginal contribution of a player to all possible coalitions, where the average is taken according to some probability distribution over the permutations of the players.

In this specific game, a random order value can be obtained in the following way. First, take all permutations of the players, and assign a probability to each permutation. Then, the random order value of player P is the expected marginal contribution of P to the value of the coalition preceding it, where the expectation is taken according to said probability distribution.

For the remainder of the paper, let us use the following notation: φ_P^n denotes the (depending on section, Shapley, weighted or random order) value of player P in the game with players $N = \{P, A_1, \ldots, A_n\}$. Then, formally,

$$\varphi_P^n = \frac{1}{(n+1)!} \sum_R \Pr(R)[v(\mathcal{P}_P^R \cup \{i\}) - v(\mathcal{P}_P^R)]$$

where R is a permutation of all players, and $\mathcal{P}_P^R$ is the set of players before P in the permutation R. [8]

Random order values are of interest for a number of reasons. First, they are a natural extension to the Shapley value and the weighted value. Furthermore, they provide a convenient framework, of which all subsequent results (Shapley value, weighted value, multiple big players) are special cases.

Utilizing the fact that the value of any coalition is zero without P, and that – conditional on player P being in a coalition – a coalition's value only depends on the number of fringe firms in it, the above expression can be simplified to the following:

$$\varphi_P^n = \frac{1}{n+1} \sum_{k=0}^{n} \Pr(|\mathcal{P}_P| = k) f(k/n),$$

where $\Pr(|\mathcal{P}_P| = k)$ is determined by the probability distribution over the permutations.

Now let us look at the asymptotic limit of this expression, as the number of fringe players goes to infinity. The following proposition demonstrates the main idea of this paper: even as the number of fringe players goes to infinity, the value of the big player does not converge to the worth of the grand coalition,

[6] This assumption is not entirely innocuous. For example, having more small firms could in theory lead to stronger competition, decreasing total profits. However, it often holds in models with network effects (Rochet and Tirole 2003) or with consumers having taste for variety (Anderson, Erkal, and Piccinin 2020).

[7] More precisely, the asymptotic value coincides with the axiomatic one derived using the uniform distribution. For a more detailed discussion, see Fogelman and Quinzii (1980).

[8] For example, if $R = \{A_1, P, A_2\}$, then $\mathcal{P}_P^R = \{A_1\}$.

$f(1)$. Or, in other words, the fringe, even when its members are infinitesimally small, still retains some bargaining power. As a corollary, the continuous fringe model can be thought of as a more tractable approximation of bargaining between a finite number of fringe players, as it retains the latter's main features.

Proposition 2. *Let f be continuous on [0, 1]. Furthermore, let us denote the random variable $\frac{|\mathcal{P}_P|}{n}$ as X_n. Assume that $X_n \xrightarrow{d} X$ with cdf $G(t)$ and (if exists) pdf $g(t)$. Then*

$$\varphi_P^\infty = \lim_{n\to\infty} \varphi_P^n = \int_0^1 f(t)\mathrm{d}G(t) = \int_0^1 g(t)f(t)\mathrm{d}t,$$

with the last equality holding if X is a continuous random variable.

Now let us turn to the value of the fringe. The following corollary shows that the aggregated value of the fringe remains positive even in the limit. In other words, even though the fringe players become smaller and more numerous, they do not lose all of their bargaining power, and still end up with a positive payoff in the end.

Corollary 1. *The aggregated value of the fringe is*

$$\varphi_A^\infty = f(1) - \int_0^1 f(t)\mathrm{d}G(t).$$

Furthermore, if f is differentiable on $[0, 1]$, then the value of the fringe can also be expressed as

$$\varphi_A^\infty = \int_0^1 G(t)f'(t)\mathrm{d}t.$$

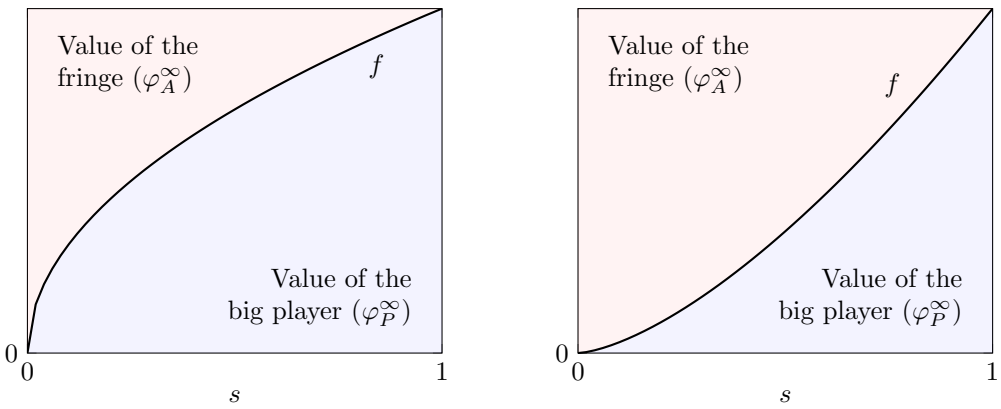
It also provides us with two ways of thinking about the value of the fringe. First, due to random order values being efficient, it can be thought of as the leftover from the total value after subtracting the value of the big player. Alternatively, as the second part of the corollary demonstrates, it is also related to the marginal contributions of the fringe agents. The only difference from the big player's value is that we have to multiply these marginal contributions by the probability of the big player appearing before the fringe firm in question, as otherwise they would be zero. The latter interpretation will be particularly useful when thinking about the value of the fringe in the multi-sided setting.

It also follows immediately from the previous proposition that (for a fixed total pie size $f(1)$) player P gets a larger share when $f(t)$ is larger for every t. That is, when only a relatively small fraction of the fringe is needed to create a large part of the total potential surplus (e.g. fringe players' products are substitutes of each other), then player P is able to appropriate a larger share of the total value. This result aligns with our natural intuition that player P has a better bargaining position in this case, than if only coalitions with most of the fringe firms on board could obtain high values. In contrast, when fringe players are more complementary to each other, the big player can only obtain a smaller share, as its bargaining power is more limited.

2.2 Special cases

2.2.1 Shapley value

Now let us turn our attention to an important special case of random order values: the Shapley value. It can be characterized by being the random order value that additionally also satisfies the axiom of symmetry. This axiom requires that if two players are identical in terms of the characteristic function, then they should receive the same value. Equivalently, it is the same as requiring that all permutations have an equal probability.



(a) Fringe players are each other's substitutes (b) Fringe players are each other's complements

Figure 1: Distribution of value between player P (red) and the fringe (blue)

The following proposition and corollary derives the limit of the Shapley value for the big player and the fringe, respectively.

Proposition 3. *Let f be continuous on [0, 1]. Then*

$$\varphi_P^\infty = \lim_{n\to\infty} \varphi_P^n = \int_0^1 f(t)\mathrm{d}t.$$

Corollary 2. *The aggregated Shapley-value of the fringe is*

$$\varphi_A^\infty = f(1) - \int_0^1 f(t)\mathrm{d}t.$$

Furthermore, if f is differentiable on $[0,1]$, then the value of the fringe can also be expressed as

$$\varphi_A^\infty = \int_0^1 tf'(t)\mathrm{d}t.$$

Because the Shapley value is a special case of the random order value, these results are quite straightforward consequences of Proposition 2 and Corollary 1. One only has to notice that each permutation having an equal probability implies that the number of firms coming before P is uniformly distributed on $1,\dots,n$, and thus the share of firms before P converges in distribution to the uniform distribution on $[0,1]$. Alternatively, a more direct proof is provided in Appendix C, or a probabilistic justification can be found in Levy and Shapley (1997).

The resulting Shapley values also have a nice geometric interpretation shown on Figure 1. Consider a rectangle with sides of length 1 and $f(1)$, respectively. The grand coalition achieves the value $f(1)$, which is also the area of the rectangle.[9] The value of the big player is then $\int_0^1 f(t)\mathrm{d}t$, which is simply the part of the rectangle below the graph of f, while the fringe gets the rest. Thus, the graph of f partitions the area (total value) into what the big player and the fringe can obtain.

Finally, the following corollary highlights that, even though the individual share of each A_i vanishes as their number goes to infinity, their total value still remains positive.

Corollary 3. *$\varphi_P^\infty < f(1)$ and $\varphi_A^\infty > 0$ for any f that is not constant.*

[9] Due to superadditivity, the grand coalition can be expected to form, and the main question is how the value is distributed between the big player and the fringe.

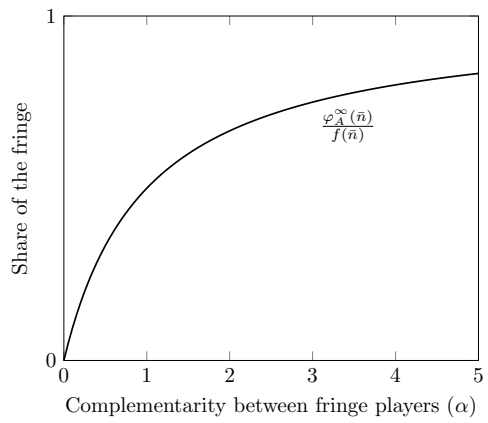


Figure 2: Example: share of the fringe as a function of α (degree of complementarity between fringe players)

Example: power function. Let us illustrate the results of this section with a simple example. Consider the case when $f(n) = n^\alpha$, for some $\alpha > 0$. Assume that the bargaining takes place between player P and a measure $\bar{n}$ of fringe players. Thus, the value that the grand coalition can achieve is $\bar{n}^\alpha$.

Proposition 11 shows that the value of player P in this case is

$$\varphi_P^\infty(\bar{n}) = \int_0^1 (s\bar{n})^\alpha = \frac{1}{\alpha+1}\bar{n}^\alpha,$$

while the fringe gets the rest, i.e.,

$$\varphi_A^\infty(\bar{n}) = \bar{n}^\alpha - \frac{1}{\alpha+1}\bar{n}^\alpha = \frac{\alpha}{\alpha+1}\bar{n}^\alpha.$$

In words, regardless of the value of $\bar{n}$, the two types of players share the total value in the same proportion. This proportion depends on the parameter α (Figure 2), which can be interpreted as the degree of substitutability between the fringe players. More specifically, the higher α is, the more complementary the fringe players are to each other, and the larger the share they obtain. In the limit, when $\alpha \to \infty$, the fringe obtains all the value, as essentially all players become indispensable, and share the total value equally. On the other hand, when $\alpha \to 0$, the big players can appropriate all the value.

2.2.2 Weighted values

This section investigates the case of players having different levels of innate bargaining power. One way to model this is by using weighted values (Shapley 1953), which are a generalization of the Shapley value. They are also a non-symmetric generalization of the Shapley value, albeit with more structure than random order values. The idea behind them is that, in addition to the set of players and the characteristic function, the game is also endowed with a weight system, which is a vector of non-negative numbers, one for each player. These weights determine how the value of a coalition is distributed amongst its members in the case when they are all equally important for the coalition. Then, relying on the linearity of the value, this distribution can be extended to the case when the players are not equally important.

These weights can be thought of as the measure of some innate bargaining power, not reflected in the value function.[10] Further support for this interpretation in certain games can be found in S. Hart and Mas-Colell (1996), demonstrating that in a certain alternating offer bargaining game, weights are related

[10] Although, this interpretation is not always appropriate. As Owen (1968) demonstrates, there are games where the weighted value is not monotone increasing in a player's weight.

to the probability of each player making the offer.[11] Additionally, Stole and Zwiebel (1996a) provides an alternative microfoundation for the weighted value in the case of non-binding contracts.

For my purposes, it is sufficient to deal with the case of simple weights. I.e., assume that there is at most one player with zero weight. For my main result, I make use of an alternative, computational characterization of the weighted values. Kalai and Samet (1987) demonstrated that weighted values can be calculated by weighting the permutations by the probabilities arising from the following sequential ordering procedure. Start with the set of all players. Let the probability of player i being the last amongst the set of remaining players R be $\lambda_i / \sum_{j \in R} \lambda_j$. Continue until all players are exhausted. This yields a well-defined probability for each permutation of players.

Now let us turn to deriving the weighted value and its limit for the big player and the fringe in this specific game. First, I propose a lemma that characterizes the distribution of the number of players before P in the limit, and then use it to derive the weighted value of the big player and the fringe. Consider the game described in the previous section, and let the weights of players of type A and P be 1 and λ, respectively. Let X_n be a random variable representing the number of players before P when players are ordered according to the previously described procedure. Then, the probability of player P having at most k players of type A before themselves is simply

$$\Pr(X_n \leq k) = \prod_{j=k+1}^{n} \frac{j}{j+\lambda}. \tag{1}$$

The following lemma establishes the continuous analogue of this statement.

Lemma 1. *As $n \to \infty$, $X_n \xrightarrow{d} X$ with the cdf $G_X(t) = \lambda^t$. Consequently, the corresponding probability density function is $g(t) = \lambda t^{\lambda-1}$.*

Now we have everything we need to derive the weighted value of both types of players. Having Equation (1) and Lemma 1 allows us to invoke Proposition 2 to obtain the following proposition.[12]

Proposition 4. *Let $f(t)$ be continuous on $[0,1]$. Then*

$$\varphi_P(\lambda, \infty) = \int_0^1 g(t) f(t) \mathrm{d}t$$

where $g(t) = \lambda t^{\lambda-1}$.

Furthermore,

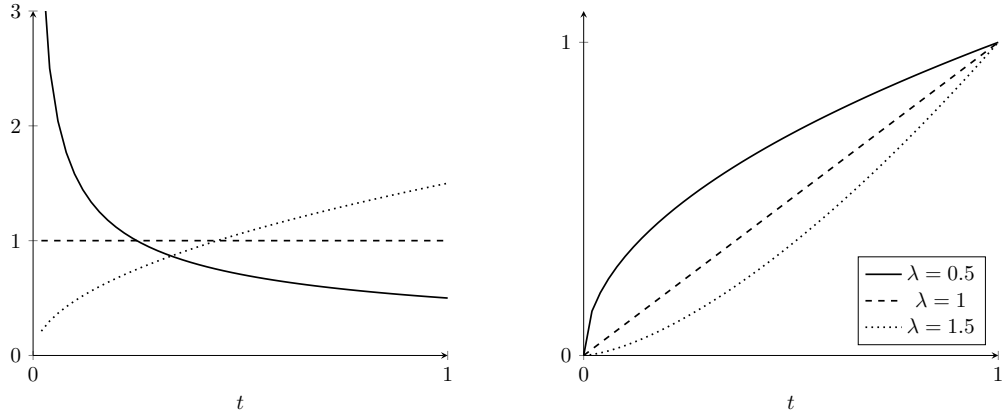
$$\begin{aligned} \varphi_A(\lambda, \infty) &= 1 - \int_0^1 g(t) f(t) \mathrm{d}t \\ &= \int_0^1 G(t) f'(t) \mathrm{d}t, \end{aligned}$$

where $G(t) = t^\lambda$.

As before, the value of the big player can still be expressed as an integral of the value function, while the value of the fringe can be expressed as an integral of the marginal contributions of the fringe players. The only difference is that, as in the case of the general random order value, the integrands are weighted by some probability distribution. Depending on the shape of this distribution, regions corresponding to different masses of fringe firms can be over or underweighted in this integral, leading to different values. Contrary to Proposition 2, the probability distribution has a specific functional form, and is characterized by a single parameter, λ. Figure 3 illustrates the weighting function for various values of λ.

[11] For example, only one player having a positive value corresponds to them making take it or leave it offers.

[12] Stole and Zwiebel (1996a) also derive essentially the same expression for the equilibrium outcome of their bargaining game with unequal profit distribution, and assert that it is equal to the weighted value.



(a) $g(t)$ (b) $G(t)$

Figure 3: Illustration of the weighting function for various values of λ in the weighted Shapley-value.

Now, let us examine how these weights impact the shares of the various players.

Corollary 4. *$\varphi_P(\lambda, \infty)$ is increasing in λ unless f is constant.*

It turns out, that a higher weight corresponds to a higher Shapley-value in this game.[13], supporting their interpretation as some kind of innate bargaining power.

Furthermore, the next corollary shows that the limit of the weighted value of P, as their weight goes to zero (infinity) is precisely the payoffs P would achieve if P were making (facing) take-it-or-leave-it offers.

Corollary 5. *If the weight of player P goes to infinity (zero), its weighted Shapley value converges to $f(1)$ ($f(0)$).*

This set of results highlights that utilizing (weighted) Shapley values to model bargaining outcomes provides an intermediate solution between inscribing all the bargaining power to one or the other player, and thus a generalization of take-it-or-leave-it offers.

2.3 Non-indispensable big player(s)

This section relaxes the assumption that the big player is indispensable. I demonstrate two ways in which the model can be extended to include non-indispensable big players, while retaining the essential structure and the tractability of the results. In the first case, there are a finite number of big players, which are perfect substitutes of each other. In the second case, there is a single big player, but the fringe is able to generate some value on its own. In both cases, the value function is still a random order value, and thus results from the main text can be applied.

2.3.1 Shapley value with multiple big players

Imagine that, instead of just one, there are m players of type P, and they are perfect substitutes of each other. That is, any coalition still needs at least one of them to create any value, but it does not matter which and how many of them are present.[14] Formally, the value function for coalition S becomes the

[13]This would not necessarily have to be the case, as demonstrated by Owen (1968).

[14]The relaxtion of this assumption, while conceptually straightforward, is not as simple to analyze. Instead of having to deal with the minimum of m independent random variables, one needs to consider all order statistics.

following:

$$v(S) = \begin{cases} 0 & \text{if } n_P(S) = 0 \\ f\left(\frac{n_A(S)}{n}\right) & \text{otherwise.} \end{cases}$$

Let us start by establishing monotonicity for this version of the model, too.

Proposition 5. *The game is monotone if f is increasing and $f(0) \geq 0$.*

On the other hand, superadditivity is not as simple in this case, as before. For example, when f is concave, then m coalitions with one platform each and the fringe divided equally amongst them clearly achieve a higher total payoff than the grand coalition. Therefore, the results of this section are most applicable to settings when multiple coalitions cannot be formed at the same time for some reason. For example, S. Hart and Mas-Colell (1996) propose an interpretation when there is a single, indivisible, non-replicable resource or technology (not captured in the value function) necessary to produce any value.

For simplicity, let us restrict our attention to probability distributions over the permutations that are symmetric with respect to the big players. That is, the probability of having k_F fringe players and k_P big players before player P_j is the same for all i.[15] As before, I assume that, as the number of fringe players goes to infinity, the distribution of the number of fringe players before player P_j to a continuous distribution.

Let us first establish the sum of the shares of all big players. Then, the value of individual big players can be obtained by relying on the symmetry assumption. The following proposition provides an expression for the limit of this total value.

Proposition 6. *Let f be continuous on [0, 1]. Let us denote the number of fringe players in any coalition S as $n_F(S)$. Furthermore, let us denote the random variable $\frac{n_F(\mathcal{P}_{P_j})}{n}$ as X_n. Assume that $X_n \xrightarrow{d} X$ with cdf $G(t)$ and (if exists) pdf $g(t)$.*

Let $H(t) = 1 - [1 - G(t)]^m$, i.e. the cdf of the minimum of the m independent random variables distributed according to G. Then

$$\varphi_P^\infty = \lim_{n\to\infty} \sum_{j=1}^{m} \varphi_{P_j}^n = \int_0^1 f(t)\mathrm{d}H(t) = \int_0^1 h(t)f(t)\mathrm{d}t,$$

with the last equality holding if X is a continuous random variable.

Notably, value in the case of multiple, substitutable big players can be interpreted as a random order value with a single big player. The probability distribution for the corresponding random order value has a specific form, and depends on the number of big players. Consequently, all the results from the previous section apply to this case as well.

As with proposition 3, this result also has a probabilistic interpretation. Let the location of the m atoms be $t_1, \ldots, t_m$, distributed independently and uniformly on the unit interval. The expected marginal contribution of t_i is only positive whenever it is the first amongst the major players. Therefore, the total value of the big players is the random order value with corresponding probability distribution being the minimum of m independent random variables describing the number of fringe players before each big player.

Due to the symmetry of the big players in terms of the permutation probabilities, the value of each individual big player is the same.

[15] This assumption can easily be relaxed, but it would complicate the results without adding much insight.

Corollary 6. *The limit of the value (as $n \to \infty$) of each player of type P is*

$$\varphi_{P_i}^{\infty,m} = \frac{1}{m}\varphi_P^{\infty} = \int_0^1 \frac{h(t)}{m} f(t)\mathrm{d}t$$

where $H(t) = 1 - [1 - G(t)]^m$, and $h(t) = H'(t)$ whenever G is differentiable.

Furthermore, the value of the fringe has a similar, marginal contribution-based interpretation as in the case of a single big player.

Corollary 7. *The per-unit Shapley-value of the fringe is*

$$\varphi_A^{\infty,m} = 1 - \int_0^1 f(t)\mathrm{d}H(t) = \int_0^1 H(t)f'(t)\mathrm{d}t,$$

where $H(t) = 1 - [1 - G(t)]^m$.

Finally, let us look at how the value of the big players and the fringe changes as a function of the number of the big players. Proposition Corollary 9 again confirms our intuition: more players means that they become more substitutable, and thus their bargaining power decreases. This not only means that the value of each individual big player decreases, but also that their total value goes down.

Corollary 8. *Let $\varphi_P^{\infty,m}$ denote the aggregated values of players of type P. $\varphi_P^{\infty,m}$ is decreasing in m and $\varphi_A^{\infty,m}$ is increasing in m unless f is constant.*

Example: Shapley value. I illustrate the effect of the number of big players on the weight function $h(t)$ by looking at the simplest random order value, the Shapley value. As shown in Proposition 3, the limit of the Shapley value in the case of a single big player is just $\int_0^1 f(t)\mathrm{d}t$. Therefore, using the notation from Proposition 2, $G(t) = t$. Conseqently, by Proposition 6, the sum of the shares of the big players corresponds to the random order value characterized by the cumulative distribution function

$$H(t) = 1 - (1 - t)^m. \tag{2}$$

That is, the value of the sum of the big players' values is given by

$$\varphi_P^{\infty,m} = \underbrace{m(1-t)^{m-1}}_{h(t)} f(t)\mathrm{d}t,$$

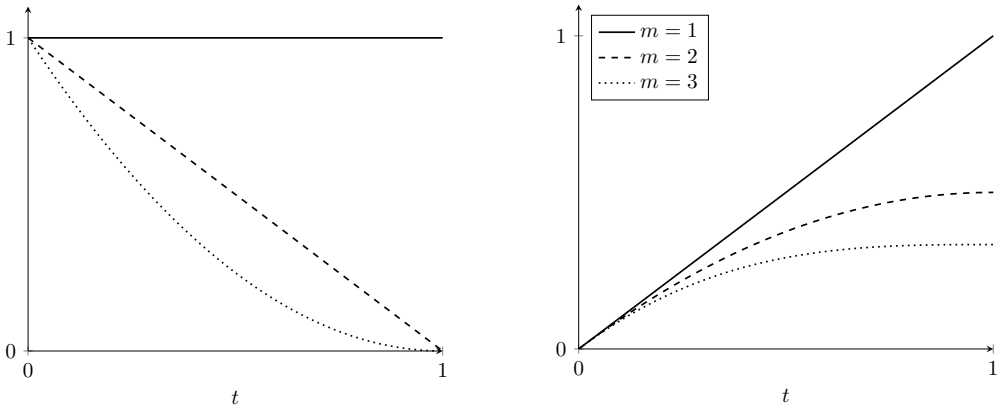
while the value that each individual big player obtains is

$$\varphi_{P_j}^{\infty,m} = \underbrace{(1-t)^{m-1}}_{h_j(t)} f(t)\mathrm{d}t.$$

Figures 4 and 5 illustrate how h_t and $h_j(t)$ depend on the number of big firms, m. In particular, it is apparent that a $H(t)$ with lower m first order stochastically dominates one with a higher number of big firms. This is the underlying reason of the big players' share decreasing in their number, i.e. Corollary 9.

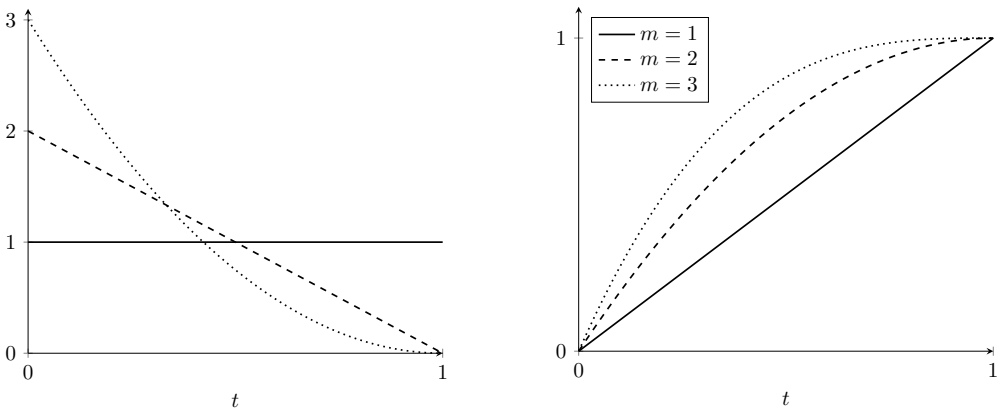
Now let us look at how the value of the big players and the fringe changes as a function of the number of the big players. Proposition Corollary 9 again confirms our intuition: more players means that they become more substitutable, and thus their bargaining power decreases. This not only means that the value of each individual big player decreases, but also that their total value goes down.

Corollary 9. *Let $\varphi_P^{\infty,m} = m\varphi_{P_i}^{\infty,m}$ denote the aggregated values of players of type P. $\varphi_P^{\infty,m}$ is decreasing in m and $\varphi_A^{\infty,m}$ is increasing in m unless f is constant.*



(a) $h_i(t)$ (b) $\int_0^t h_i(s)\mathrm{d}s$

Figure 4: Illustration of the weighting function for various values of m in the multiple big player case – individual big players.



(a) $h(t)$ (b) $H(t) = \int_0^t h(s)\mathrm{d}s$

Figure 5: Illustration of the weighting function for various values of m in the multiple big player case – total of big players.

Finally, let us show that the value of the fringe has a similar, marginal contribution-based interpretation as in the case of a single big player.

Corollary 10. *The per-unit Shapley-value of the fringe is*

$$\varphi_A^{\infty,m} = 1 - \int_0^1 f(t)\mathrm{d}G(t) = \int_0^1 G(t)f'(t)\mathrm{d}t,$$

where $G(t) = 1 - (1-t)^m$.

2.3.2 Small players can generate some value on their own

Another way of relaxing the indispensability of the big player is assuming that the small players can generate some value on their own. Suppose that the big player still provides some value to any coalition, but a coalition of fringe players can achieve a positive value even without it. Formally, the value function

is now the following:

$$v(S) = \begin{cases} f_0\left(\frac{n_A(S)}{n}\right) & P \notin S \\ f\left(\frac{n_A(S)}{n}\right) & \text{otherwise.} \end{cases}$$

As in the case of multiple big players, monotonicity is straightforward to establish.

Proposition 7. *The game is monotone if f is increasing and $f(t) \geq f_0(t) \geq 0 \forall t$.*

Also similarly, conditions for superadditivity are not as simple. Therefore, the simultaneous formation of multiple coalitions must be excluded in this setting as well.

On the other hand, the random order values and their limits are very straightforward to characterize. The value of the big player is still its average marginal contribution. The only difference is that this marginal contribution (conditional on s share of the fringe coming before P) is not $f(s)$, but rather $f(s) - f_0(s)$. This is because the big player is not indispensable, as s mass of fringe firms can achieve $f_0(s)$ even without it.

Proposition 8. *Let f be continuous on [0, 1]. Furthermore, let us denote the random variable $\frac{|\mathcal{P}_P|}{n}$ as X_n. Assume that $X_n \xrightarrow{d} X$ with cdf $G_X(t)$ and (if exists) pdf $g(t)$. Then*

$$\varphi_P^\infty = \lim_{n\to\infty} \varphi_P^n = \int_0^1 f(t)\mathrm{d}G(t) = \int_0^1 g(t)[f(t) - f_0(t)]\mathrm{d}t,$$

with the last equality holding if X is a continuous random variable.

The value of the fringe is also similar to the baseline case, except now fringe players have a non-zero marginal contribution even when P is not present. Therefore, there is an additional term in the integral, corresponding to the marginal value of the fringe when P is not present.

Corollary 11. *The aggregated value of the fringe is*

$$\varphi_A^\infty = f(1) - \int_0^1 f(t) - f_0(t)\mathrm{d}G(t).$$

Furthermore, if f and f_0 are differentiable on $[0, 1]$, then the value of the fringe can also be expressed as

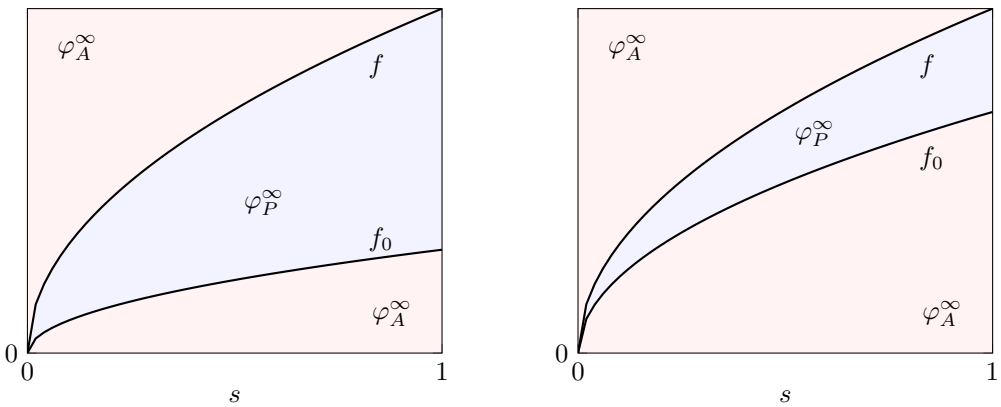
$$\varphi_A^\infty = \int_0^1 G(t)f'(t) + (1 - G(t))f_0'(t)\mathrm{d}t.$$

It is clear that the value of the big player is decreasing in f_0 (Figure 6). Consequently, the value of the fringe is increasing in f_0, which is essentially its outside option (what it can achieve without the big player). This is a natural result, and again, it is in line with what one would expect from bargaining theory.

3 Heterogeneous fringe

Up until this point, we have only considered games with a single type of small player. Now imagine that, in addition to player P, there are L different varieties of smaller players: $\{A_1^l, \ldots, A_n^l\}$ where $1 \leq l \leq L$. The idea is the same as before: I assume that P is necessary for any coalition to have a positive value, and the small players are identical to each other *within their types*. The substitutability of the different types of small players will be captured by the – now multivariate – function f.

Let us now turn to the formal definition of the game. For a finite number of fringe players, the coalitional form of the game (N, v) is the following. The set of players is $N = \{P, A_1^1, \ldots, A_n^1, \ldots, A_1^L, \ldots, A_n^L\}$,



(a) Fringe can achieve relatively little without P (b) Fringe can achieve relatively more without P

Figure 6: Distribution of value between player P (blue) and the fringe (red)

which in total contains $nL + 1$ players.[16] The characteristic function is

$$v(S) = \begin{cases} 0 & \text{if } P \notin S \\ f\left(\frac{n_{A^1}(S)}{n}, \dots, \frac{n_{A^L}(S)}{n}\right) & \text{otherwise,} \end{cases}$$

where f is now a multivariate function from $[0,1]^L$ to $\mathbb{R}$.

A prominent interpretation of the two-sided version of this model would be a platform marketplace with a set of sellers and a set of buyers, where all three sides possess some amount of bargaining power. However, player P does not have to be in the "middle" of the transactions for this framework to be applicable. For example, it also captures the situation of a single upstream producer, a large number of downstream firms, and a similarly large number of customers, as long as both the customers and the downstream firms can participate in the bargaining process.[17] More than two types of fringe players can be used to model a platform with more than two sides, such as a platform with both buyers and sellers, and a set of advertisers, or, following Stole and Zwiebel (1996a), bargaining between a firm and multiple types of input suppliers.

For the subseuent discussion, let us follow the structure of the previous section. I start by characterizing conditions for monotonicity and superadditivity to support the bargaining interpretation and the formation of the grand coalition. It turns out that necessary and sufficient conditions for these properties are the straightforward analogues of those from the one-sided case.

Proposition 9. *The game (N, v) is monotone and superadditive if and only if f is increasing in all of its arguments.*

The intuition for this is the same as before: superadditivity is equivalent to monotonicity due to the fact that no coalition can have a positive value without P, and monotonicity boils down to f being increasing in all of its arguments.

In the next subsections, I will first examine random order values, and then two important special cases: the Shapley value and the weighted value. Many of the results will be similar to the one-sided case, including the marginal contribution interpretation of the value of the fringe, and the geometric interpretation of the Shapley value. Furthermore, it turns out that even though the number of fringe

[16] In this formulation, we assume that the number of players of each type is the same. However, this is not as restrictive as it might seem. In the limit, each type of player will become infinitesimally small, and the exact number of players of each type will not matter.

[17] A setting where these assumptions might be plausible is, for example, the new car market with a car producer, a number of independent dealerships, and customers who might engage in some form of negotiation.

players is a multi-dimensional object in this case, for the limit of the Shapley and weighted values, one only has to integrate over a one-dimensional manifold within the unit hypercube, which makes the results more tractable.

3.1 Random order values

First, let us examine the most general case: the random order value. Let us start by establishing the analog of Proposition 2 for the many-sided case to obtain the limit of the value of the big player.

Proposition 10. *Let f be continuous on $[0,1]^L$. Furthermore, let us denote the random vector of the proportions of the various types of fringe players before P as X_n:*

$$X_n = \left(\frac{n_{A_1}(\mathcal{P}_P)}{n}, \dots, \frac{n_{A_L}(\mathcal{P}_P)}{n} \right).$$

Assume that $X_n \xrightarrow{d} X$ with cdf $G_X(t_1, \dots, t_L)$ and (if exists) pdf $g(t_1, \dots, t_L)$. Then

$$\begin{aligned} \varphi_P^\infty = \lim_{n\to\infty} \varphi_P^n &= \int_0^1 \cdots \int_0^1 f(t_1, \dots, t_L) \mathrm{d}G(t_1, \dots t_L) \\ &= \int_0^1 \cdots \int_0^1 g(t_1, \dots, t_n) f(t_1, \dots, t_L) \mathrm{d}t_1 \dots \mathrm{d}t_L, \end{aligned}$$

with the last equality holding if X is a continuous random variable.

The proposition above is almost identical to Proposition 2, with the only difference being that the integral is now over the L-dimensional unit hypercube, due to f being a function of L variables.

While the above proposition is very general in terms of permutation probabilities, it is not very tractable in practice, as one has to integrate over the entire, L-dimensional domain of f. It would be much more convenient, if one only had to be concerned with a one-dimensional manifold within the hypercube instead. It turns out, that for a number of important cases (namely, the Shapley value and the weighted value) it is indeed the case.

Let us start by establishing a general result, after which I will show that the Shapley value and the weighted value are special cases of it. The following lemma provides sufficient conditions, under which the limit of the value of the big player can be expressed as an integral over a one-dimensional path, even though the value for any finite number of players depends on the whole domain of f.

Lemma 2. *Assume that $X_n \xrightarrow{d} X$ where X is a degenerate distribution in the following sense: $\exists\, a_l : [0,1] \to [0,1], l = 1, \dots, L$, such that*

$$X = (a_1(\xi), \dots, a_L(\xi)),$$

where ξ is a random variable on $[0,1]$ with cdf $H(s)$. In words, the whole probability mass is concentrated on the manifold $(a_1(s), \dots a_L(s)), t \in [0,1]$. Then

$$\varphi_P^\infty = \lim_{n\to\infty} \varphi_P^n = \int_0^1 f(a_1(s), \dots a_L(s)) \mathrm{d}H(s).$$

Furthermore, if f's partial derivatives exist on $\{(a_1(t), \dots, a_L(t)) : t \in [0,1]\}$ the value of fringe l has the following limit:

$$\varphi_{A^l}^\infty = \lim_{n\to\infty} \varphi_{A^l}^n = \int_0^1 H(s) a_l'(s) \partial_l f(a_1(s), \dots a_L(s)) \mathrm{d}s.$$

The lemma states, that if the permutation probabilities are such that the number of fringe players before P converges to a degenerate distribution, then one only has to be concerned with just a tiny fraction of the production function f. Namely, we only need to know how the function behaves on the set $\{(a_1(t), \ldots, a_L(t)) : t \in [0,1]\} \subset [0,1]^L$.[18] This result is essentially a generalization of the diagonal formula from Aumann and Shapley (2015) or Stole and Zwiebel (1996a) to the case where permutation probabilities are not uniform. The intuition, however, is the same law-of-large-numbers-type argument: if the number of fringe players is large, then it is very unlikely that, after a random ordering, their proportion is very different from its expected value. The main difference is that, in this case, this conditional expectations are not linear, but rather given by the a_l functions.

In the remainder of this section, I will show that the Shapley value and the weighted value satisfy the assumptions of the above lemma, and thus the limit of those values can be expressed as an integral over a well-defined one-dimensional path.

3.2 Special cases

3.2.1 Shapley-value

Let us now turn our attention to the Shapley value. As hinted before, due to the uniformity of the permutation probabilities, we get a diagonal formula in this case. Proposition 11 establishes this result formally.

Proposition 11. *Let f be continuous on $[0,1]^L$. Then,*

$$\varphi_P^\infty = \int_0^1 f(t, \ldots, t)\mathrm{d}t$$

and

$$\varphi_{A^l}^\infty = \int_0^1 t\partial_l f(t, \ldots, t)\mathrm{d}t.$$

That is, in the case of the Shapley value, one has to integrate over the diagonal of the unit hypercube to obtain the limit of the value of the big player.[19] Figure 7 illustrates this result in the case of two types of fringe players: the value of player P is the integral of f over the diagonal of the unit square. Furthermore, this result also demonstrates how the rest of the value is divided amongst the fringe players. Namely, those having higher marginal contributions along the diagonal get a larger share of the value.

3.2.2 Weighted value

The final part of this section is dedicated to the weighted value in the many-sided case. The set of players and the characteristic function are the same as in the previous section, but, as before, each player has a weight attached to them. I will assume that fringe players are identical to each other within their types also with respect to their weights. That is, the weight system for the game is the following: player P has weight λ_P, while players A_i^l have weight λ_l.

I will show, that as with non-weighted Shapley value, the limit of the weighted value can be expressed as an integral over a one-dimensional path in the weighted case as well. However, now this path is not the diagonal of the unit hypercube, but a function of the fringe players' weights. The reason for this is that not all permutations have the same probability of occurring: those with players having higher weights ordered later are more likely to occur.

[18] For the fringe players, we also need the partial derivatives. Therefore, technically, the behavior of f on some arbitrarily small neighborhood of the set also matters.

[19] Stole and Zwiebel (1996a) also obtain the expression for the big player's value for the limiting case of their bargaining game. The proof of this proposition demonstrates a way to obtain it as a limit of the Shapley value of the finite games.

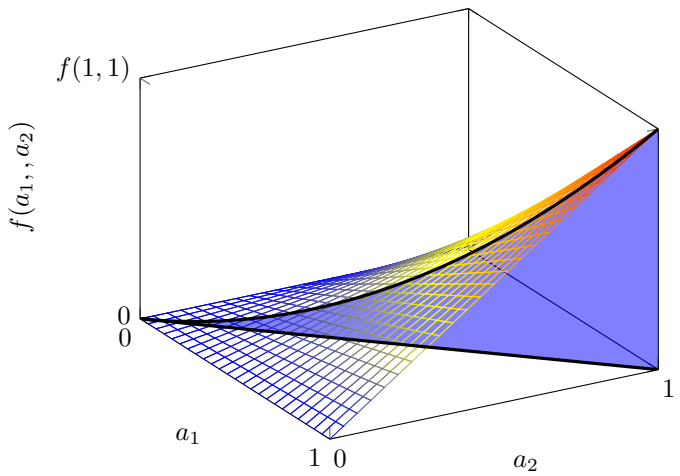


Figure 7: Illustration of the Shapley value in the case of two types of fringe players. The blue area corresponds to the value of the platform (with a scaling factor of $\sqrt{2}$ to account for the length of the diagonal)

The following statement is the central theorem of this paper. In fact, most of the propositions in the previous sections can be considered as special cases of it.

Theorem 1. *Let f be continuous on $[0,1]^L$. Then,*

$$\varphi_P^\infty = \int_0^1 \lambda_P t^{\lambda_P - 1} f(t^{\lambda_1}, \dots, t^{\lambda_L}) \mathrm{d}t$$

and

$$\varphi_{A^l}^\infty = \int_0^1 t^{\lambda_P} \lambda_l t^{\lambda_l - 1} \partial_l f(t^{\lambda_1}, \dots, t^{-L}) \mathrm{d}t.$$

The intuition is the following. A higher weight for a player means that it has a higher chance of being relatively late in a random permutation. Therefore, conditional on s proportion of the fringe coming before P, the various types' proportions do not reflect their population shares $(1/L)$. For any $s \in (0,1)$, the proportion of fringe players with a higher weight is lower than their population share, while the proportion of fringe players with a lower weight is higher than their population share.

Figure 8 illustrates this phenomenon in the case of two types of fringe players. Fringe players of type A^2 have a higher weight than those of type A^1, and thus the path the integral is taken over contains more of the latter than the former.

Finally, player P's probability of coming after s proportion of the fringe is also not uniform, but rather influenced by the relationship between its own and the fringe players' weights.[20] Therefore, the integral is not taken with respect to the uniform distribution, but rather one resembling that of Proposition 4. The result is the following: when the big player has a higher weight, its chances of being relatively late in the ordering are higher, and thus – due to f being monotone increasing – its value is higher. The following, simple corollary formalizes this idea.

Corollary 12. *φ_P^∞ is increasing in λ_P unless f is constant.*

This result is analogous to Corollary 4, and demonstrates that a higher weight for player P corresponds to a higher value in this game as well.

[20]This is the same mechanism as in the case of the one-sided weighted value.

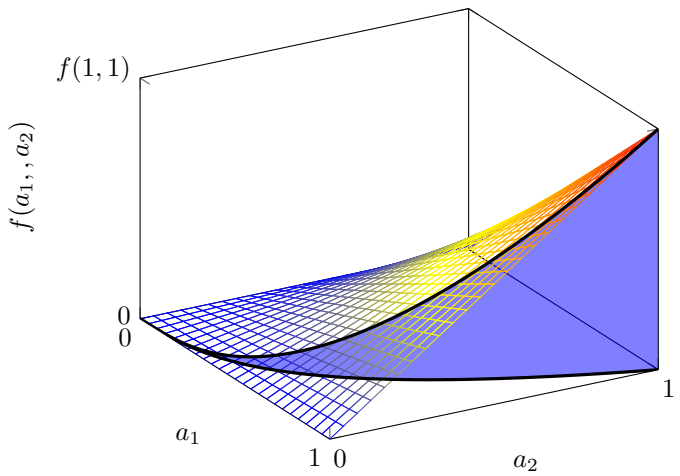


Figure 8: Illustration of the weighted value in the case of two types of fringe players. Players of type A^2 have a higher weight than those of type A^1, thus the integral is not taken over the diagonal. The blue area corresponds to the value of the platform (the area have to be scaled so that the length of the path integrated over is one)

4 Example application

In this section, I apply the ideas from the previous sections to a simple model of two-sided platforms. I follow the modelling approach of Armstrong (2006), focusing on the monopoly platform case. The model presented in this section expands on the original model in two main ways. First, instead of assuming that each player's utility is linear in the number of the other side's players, I allow for more general non-linear network effects. This extension lets me capture and vary the substitutability of the two sides. Second, and most importantly, I add bargaining between the platform and the two sides. That is, besides the welfare-maximizing and unilateral price setting cases presented in Armstrong (2006), I also consider a case where the platform and the two sides bargain over the entry fees. I assume that the outcomes of this bargaining process are described by the weighted value.

4.1 Model

Imagine a two-sided market with a continuum of players on both sides and a single platform connecting those two sides. Furthermore, assume that the utility that a player on side i derives from participating in the market is an increasing function[21] of the number of players on the other side (n_j), minus the entry fee charged by the platform (p_i):

$$u_i = \alpha_i n_j^{\gamma_i} - p_i.$$

In this formulation, $\alpha \geq 0$ and $\gamma > 0$ determine the strength and shape of the network effects, respectively.

Following Armstrong (2006), let us model player entry also in a reduced form manner. Assume that there is a continuum of potential entrants on both sides of the market. The number of actual entrants depends on the utility they would derive from participating in the market.[22] Let us denote this

[21] Armstrong (2006) – along with much of the early literature on two-sided markets (e.g. Rochet and Tirole 2003; Hagiu 2006) – assume that the utility of a player is linear in the number of players on the other side. Using a more general function allows me to model different degrees of network effects.

[22] Such assumptions are usually justified by some idiosyncratic cost of entry, the distribution of which determines the function ϕ. For simplicity, I refrain from explicitly modeling this cost in the main text, but Appendix A provides a discussion of how one can incorporate it into the model, its impact on bargaining outcomes.

relationship as $\phi_i(u_i)$, where ϕ_i is a strictly increasing, differentiable function. Then,

$$n_i = \phi_i(u_i).$$

Finally, let us assume that the platform can charge lump-sum entry fees to both sides. These fees can also be negative, in which case they are interpreted as subsidies. The platform's profit is the sum of entry fees, minus the cost of serving the entrants (F_i for each entrant):

$$\pi(n_1, n_2) = n_1 p_1 + n_2 p_2 - F_1 n_1 - F_2 n_2.$$

I consider two cases as to the determination of the entry fees, as well as a welfare-maximizing benchmark. By the latter, I mean the entry fees that maximize social welfare, i.e. the sum of the utilities of the two sides plus the platform's profit. After that, I first consider a model analogous to that in Armstrong (2006), where the platform can unilaterally commit to any entry fees it wishes. Finally, I examine and contrast the case where the entry fees the result of a bargaining process in such a way that the resulting utilities and profits correspond to the various players' (weighted) values.

4.2 Welfare-maximizing entry fees

Let us first consider what the entry fees would be if they were set to maximize social welfare. Instead of approaching this problem directly, I rely on the following observation: if entry fees correspond to the externality that a player's entry imposes on the other players, then the resulting entry decisions are welfare-maximizing.

In this specific case, there are two externalities to consider. First, it costs the platform F_i to serve an additional player on side i. Second, the entry of a player on side i increases the utility on side j due to the presence of network effects. Welfare is maximized when the price reflects the balance of these two factors.

Proposition 12. *The welfare-maximizing entry fees for side i are characterized by*

$$p_i^* = F_i - \alpha_j \gamma_j n_j n_i^{\gamma_j - 1}.$$

As in Armstrong (2006), for any $\alpha_j > 0$ welfare-maximizing F_i is below the platform's cost of serving player i. Thus, the platform's profit would necessarily be negative. Therefore, while this case is useful as a benchmark, it is not particularly realistic when the platform has a say in determining entry fees, or even the possibility to exit the market.

As an immediate corollary, utilities in this case are given by

$$u_i^*(n_i, n_j) = \alpha_i n_j^{\gamma_i} + \alpha_j \gamma_j n_j n_i^{\gamma_j - 1} - F_i$$

for each individual player on side i. Consequently, the total welfare on side i, as it is simply the number of entrants times the non-idiosyncratic utility of its players, plus the sum (integral) of the idiosyncratic utilities.

$$W_i^*(n_i, n_j) = \alpha_i n_i n_j^{\gamma_i} + \gamma_j \alpha_j n_j n_i^{\gamma_j} - n_i F_i. \tag{3}$$

That is, players on side i obtain the total generated by themselves, plus γ_j times the total utility generated by the other side, net of the cost of serving them. The utility of side i depends positively on both α_i (mechanically) and α_j (a stronger positive externality on the other players implies a lower entry fee and thus higher utility). Furthermore, it is increasing in the number of entrants on the other side through two channels: (1) entrants on the other side generate utility for side i through network effects, and (2) players on side i are compensated for the positive externality they impose on the other side.

The effects of γ_j are somewhat more complex, and are best understood through the dependence of the utility on side i on n_i. Total welfare on side i is always increasing in n_i, but individual utilities are not. If $\gamma_j > 1$ (i.e., network effects are convex), then u_i is indeed increasing in n_i, as the positive externality each individual player imposes on the other side is increasing in the number of entrants. Conversely, when $\gamma_j < 1$, the positive externality of the whole side is still increasing in their number (therefore, W_i is increasing in n_i), but the positive externality each individual player imposes is now a decreasing function of it (thus, u_i is decreasing in n_i).

4.3 Platform sets prices unilaterally

Now let us examine what happens when the platform sets the entry fees unilaterally. First, rewrite the platform's profit as a function of utilities, rather than the number of entrants:

$$\pi_P(u_1, u_2) = \phi_1(u_1)[\alpha_1\phi_2(u_2)^{\gamma_1} - u_1 - F_1] + \phi_2(u_2)[\alpha_2\phi_1(u_1)^{\gamma_2} - u_2 - F_2].$$

If the platform can set entry fees unilaterally, then it's problem is to choose u_1 and u_2 to maximize this expression.

Proposition 13. *Assume that $\pi_P(u_1, u_2)$ is concave in u_i and u_2. Then, the profit-maximizing entry fees for side i are characterized by*

$$\begin{aligned} p_i^u &= F_i - \alpha_j\gamma_j n_j n_i^{\gamma_j - 1} + \frac{\phi_1(u_1)}{\phi_1'(u_1)} \\ &= p_i^* + \frac{\phi_1(u_1)}{\phi_1'(u_1)}. \end{aligned}$$

As one would expect, profit-maximizing entry fees are higher than welfare-maximizing ones. In addition to the comparative statics of the welfare-maximizing case, the unilaterally chosen also depend on the elasticity of entry. When this elasticity is low, the can set high entry fees, as the reduction in the number of entrants is not relatively minor.

For ease of comparison with the bargaining case, let us also derive the utilities and welfare on side i in this case, too:

$$\begin{aligned} u_i^u(n_1, n_2) &= \alpha_i n_j^{\gamma_i} + \alpha_j\gamma_j n_j n_i^{\gamma_j - 1} - F_i - \frac{\phi_1(u_i)}{\phi_1'(u_i)} \\ W_i^u(n_1, n_2) &= \alpha_i n_i n_j^{\gamma_i} + \gamma_j\alpha_j n_j n_i^{\gamma_j} - n_i F_i - n_i^2(\phi_i^{-1})'(n_i). \end{aligned}$$

Clearly, equilibrium utilities must be smaller than in the welfare-maximizing case. Due to ϕ_i being increasing, the number of entrants is also lower.

4.4 Bargaining

Finally, let us consider the case when the platform and the entrants bargain over the value they generate. Assume that n_1 and n_2 players have decided to enter on sides 1 and 2, respectively. Furthermore, assume that players' bargaining weights are λ_1, λ_2 and λ_P.

The characteristic function of the game takes a similar form to that in Section 3. If the platform is not a member of a coalition, then the value that coalition can generate is zero. Otherwise, it is simply the sum of the utilities of the two sides minus the cost of serving them.[23] This value, as a function of the

[23] Entry fees paid to the platform do not matter, as they are within-coalition transfers, and thus do not affect the total value.

number of entrants, is

$$w(n_1, n_2) = n_1\alpha_1 n_2^{\gamma_1} + n_2\alpha_2 n_1^{\gamma_2} - F_1 n_1 - F_2 n_2.$$

The share received by each type of player is then easily established using Theorem 1.

Proposition 14. *The weighted values of the various sides are*

$$\pi_P^b(n_1, n_2) = \underbrace{\frac{\lambda_P}{\lambda_P + \lambda_1 + \lambda_2\gamma_1}}_{:=S_P^{b_1}} \alpha_1 n_1 n_2^{\gamma_1} + \underbrace{\frac{\lambda_P}{\lambda_P + \lambda_2 + \lambda_1\gamma_2}}_{:=S_P^{b_2}} \alpha_2 n_2 n_1^{\gamma_2} - \underbrace{\frac{\lambda_P}{\lambda_P + \lambda_1}}_{:=S_P^{c_1}} n_1 F_1 - \underbrace{\frac{\lambda_P}{\lambda_P + \lambda_2}}_{:=S_P^{c_2}} n_2 F_2$$

$$W_1^b(n_1, n_2) = \underbrace{\frac{\lambda_1}{\lambda_P + \lambda_1 + \lambda_2\gamma_1}}_{:=S_1^{b_1}} \alpha_1 n_1 n_2^{\gamma_1} + \underbrace{\frac{\lambda_1\gamma_2}{\lambda_P + \lambda_2 + \lambda_1\gamma_2}}_{:=S_1^{b_2}} \alpha_2 n_2 n_1^{\gamma_2} - \underbrace{\frac{\lambda_1}{\lambda_P + \lambda_1}}_{:=S_1^{c_1}} n_1 F_1$$

$$W_2^b(n_1, n_2) = \underbrace{\frac{\lambda_2\gamma_1}{\lambda_P + \lambda_1 + \lambda_2\gamma_1}}_{:=S_2^{b_1}} \alpha_1 n_1 n_2^{\gamma_1} + \underbrace{\frac{\lambda_2}{\lambda_P + \lambda_2 + \lambda_1\gamma_2}}_{:=S_2^{b_1}} \alpha_2 n_2 n_1^{\gamma_2} - \underbrace{\frac{\lambda_2}{\lambda_P + \lambda_2}}_{:=S_2^{c_2}} n_2 F_2.$$

Total welfare $w(n_1, n_2)$ can be divided into four parts: utility generated by side 1 and 2, and the costs of serving side 1 and 2. Each side's share of these parts is given by $S_i^{b_1}, S_i^{b_2}, S_i^{c_1}$ and $S_i^{c_2}$, respectively. As it turns out, these shares only depend on the bargaining weights $(\lambda_P, \lambda_1, \lambda_2)$ and the network effects (γ_1, γ_2), but not on the actual number of entrants.[24] Furthermore, all of these shares are increasing in one's own bargaining weight, and decreasing in the others' weights. Finally, and perhaps most interestingly, the share of side i from the value generated on side j is also increasing in γ_j. This means that if players on side i are less substitutable (or, equivalently, more complementary) in terms of contributing to the utility of the other side, then they will receive a larger share of the value generated there.

These utilities are remarkably similar in structure to those obtained in the welfare maximizing case. The main difference is that each part (both costs and positive utilities) is reweighted, and side i only receives a fraction of it. It is also worth noting that $W_i^b(n_1, n_2) < W_i^*(n_1, n_2)$, and, as a consequence, entry is lower than what would be socially optimal.[25] These similarities and differences are examined in more detail in the next subsection.

For better comparison, let us also derive implied entry fees and utilities.[26] The latter is simply average welfare, i.e.:

$$u_i^b(n_1, n_2) = \frac{\lambda_i}{\lambda_P + \lambda_i + \lambda_j\gamma_i}\alpha_i n_j^{\gamma_i} + \frac{\lambda_i\gamma_j}{\lambda_P + \lambda_j + \lambda_i\gamma_j}\alpha_j n_j n_i^{\gamma_j - 1} - \frac{\lambda_i}{\lambda_P + \lambda_i}F_i.$$

Finally, the implied entry fees can be calculated from the equation $u_i = \alpha_i n_j^{\gamma_i} - p_i$, giving

$$p_i^b(n_1, n_2) = \frac{\lambda_i}{\lambda_P + \lambda_i}F_i - \frac{\lambda_i\gamma_j}{\lambda_P + \lambda_j + \lambda_i\gamma_j}\alpha_j n_j n_i^{\gamma_j - 1} + \frac{\lambda_P + \lambda_j\gamma_i}{\lambda_P + \lambda_i + \lambda_j\gamma_i}\alpha_i n_j^{\gamma_i}.$$

4.5 Comparison

Let us now compare the outcomes of the three cases described in the previous subsections. To highlight the similarity to Armstrong (2006), I will first consider the case of linear network effects. Furthermore,

[24] This is due to the characteristic function being a polynomial of the number of entrants. Just like in Section 2.2.1, each constituent of the polynomial are divided in a constant proportion.

[25] The easiest way to see it is that in the welfare-maximizing case the platform's profit is negative, while in the bargaining case it is positive. Therefore, fringe utilities must be lower in the latter case.

[26] If one wished to close the model, the number of entrants in equilibrium could be obtained by solving the system of equations $n_i = \phi_i(u_i(n_1, n_2)), n \in \{1, 2\}$, just like in the other cases.

for simplicity of exposition, I start by examining the non-weighted Shapley value before moving on to the weighted value.

Assume that network effects are linear, i.e. $\gamma_i = 1$ for $i \in \{1, 2\}$. Also, focus on non-weighted Shapley values ($\lambda_P = \lambda_1 = \lambda_2 = 1$). From the previous results, the entry fees for side i are

$$\begin{aligned} p_i^* &= \underbrace{F_i}_{(1)} - \underbrace{\alpha_j n_j n_i}_{(2)}, \\ p_i^u &= \underbrace{F_i}_{(1)} - \underbrace{\alpha_j n_j n_i}_{(2)} + \underbrace{n_i (\phi_i^{-1})'(n_i)}_{(3)} \\ p_i^b &= \underbrace{\frac{1}{2} F_i}_{(1)} - \underbrace{\frac{1}{3} \alpha_j n_j n_i}_{(2)} + \underbrace{\frac{2}{3} \alpha_i n_j}_{(4)}, \end{aligned}$$

for the welfare-maximizing, unilateral price setting and bargaining cases, respectively.

All of these expressions bear some similarities to each other in that the price is somehow related to the externalities that an entrant imposes on the other players. In fact, the welfare-maximizing price is just that: the cost of serving one additional player (1) and the positive externality imposed on the other side (2). The unilateral pricing case also includes these two factors in full, but also adds a third factor, related to the elasticity of entry (3). Namely, the platform extracts some of the surplus generated by the entrants, and the surplus it wants to extract depends on the elasticity of entry, $\phi_i'(u_i)$. It is the same mechanism that leads to markups in a simple monopolistic pricing model.

The bargaining case also includes the first two factors, albeit they are somewhat different. As opposed to the first two cases, the price only contains some fraction of the externalities, while the rest is shared with the platform and the other side.[27] Furthermore, some part of the welfare generated on side i is also shared with the other players, and thus appears in the price (4). This is in contrast to both the welfare-maximizing and unilateral pricing cases, where the entry fee did not depend on the actual utility that was materialized on the given side. Therefore, bargaining introduces some interesting departures from both the welfare-maximizing and unilateral pricing cases.

Now let us consider non-linear network effects. The welfare-maximizing, unilateral pricing and bargaining entry fees are as follows:

$$\begin{aligned} p_i^* &= \underbrace{F_i}_{(1)} - \underbrace{\alpha_j \gamma_j n_j n_i^{\gamma_j - 1}}_{(2)}, \\ p_i^u &= \underbrace{F_i}_{(1)} - \underbrace{\alpha_j \gamma_j n_j n_i^{\gamma_j - 1}}_{(2)} + \underbrace{n_i \kappa_i'(n_i)}_{(3)} \\ p_i^b &= \underbrace{\frac{1}{2} F_i}_{(1)} - \underbrace{\frac{\gamma_j}{2 + \gamma_j} \alpha_j n_j n_i^{\gamma_j - 1}}_{(2)} + \underbrace{\frac{1 + \gamma_i}{2 + \gamma_i} \alpha_i n_j^{\gamma_i}}_{(4)}. \end{aligned}$$

For the most part, the intuition is the same as in the linear case. In fact, in the welfare-maximizing and unilateral pricing cases, nothing changes apart from the marginal externality imposed on the other side (2) being slightly different. On the other hand, there is one important difference in the bargaining case. In the linear case, the terms related to the value generated on each side were shared between the players in a constant proportion, independent of the model parameters. In this non-linear case, however, these shares depend on the shape of the network effects. In particular, when side i's network externality on side j is more convex (e.g. firms on side i are less substitutable), then it receives a larger share of the

[27] Due to the linearity of the model and the symmetry of the Shapley value, each part of the value is shared equally between players who participate in producing them. This is the reason why, for example, the platform and the firms share the cost of serving the entrants (1) equally, or why all three types of players share the value generated on a given side (2) equally.

value generated on the other side. This is signified by the fact that the coefficient of side i's network externality, $\frac{\gamma_j}{2+\gamma_j}$, is increasing in γ_j. This result demonstrates that bargaining outcomes are not just a split of the value, but also take into account the shape of the network effects.

Finally, for completeness, let us also consider the case of weighted values. The following theorem summarizes the entry fees in this most general case.

Theorem 2. *Let p_i^*, p_i^u and p_i^b denote the welfare-maximizing, unilateral pricing and bargaining entry fees, respectively. Then,*

$$p_i^* = \underbrace{F_i}_{(1)} - \underbrace{\alpha_j \gamma_j n_j n_i^{\gamma_j - 1}}_{(2)},$$

$$p_i^u = \underbrace{F_i}_{(1)} - \underbrace{\alpha_j \gamma_j n_j n_i^{\gamma_j - 1}}_{(2)} + \underbrace{n_i \kappa_i'(n_i)}_{(3)}$$

$$p_i^b = \underbrace{\frac{\lambda_i}{\lambda_P + \lambda_i} F_i}_{(1)} - \underbrace{\frac{\lambda_i \gamma_j}{\lambda_P + \lambda_j + \lambda_i \gamma_j} \alpha_j n_j n_i^{\gamma_j - 1}}_{(2)} + \underbrace{\frac{\lambda_P + \lambda_j \gamma_i}{\lambda_P + \lambda_i + \lambda_j \gamma_i} \alpha_i n_j^{\gamma_i}}_{(4)}.$$

In the case of the cost of serving one additional player (1), the addition of bargaining weights simply leads to players sharing the cost in a constant proportion, according to their weights.[28] The higher the weight, the larger the share for any given player. While the latter observation is true for non-linear terms, as well, the influence of the weights is more complex for those. Namely, it also depends on the shape of value function. For example, in term (2), λ_i interacts with the parameter determining the shape of the network effects, γ_j Therefore, the weighted value is more than just a simple reweighting of the non-weighted value, and its effects can be quite complex depending on the assumptions on the value function.

Whether bargaining leads to a higher or lower total welfare depends on the parameters of the model. For example, if λ_P is very high, then $p_i^b > p_i^u > p_i^*$, leading to fewer firms entering and lower total surplus not only than the welfare-maximizing case, but also than the unilateral pricing case. That is, it can happen that everybody looses from bargaining.[29] On the other hand, if λ_P is sufficiently low, then one can find weights λ_1, λ_2, such that $p_i^u > p_i^b > p_i^*$. In such a case, the platform would like to set higher entry fees than what it can achieve via bargaining for both sides. Furthermore, the total welfare under bargaining is between the welfare-maximizing and unilateral pricing cases. Therefore, whether bargaining leads to higher or lower welfare than unilateral pricing depends on model parameters. The platform, however, always achieves a (weakly) lower profit than if it could set the entry fees unilaterally.

Finally, it is always true that $p_1^b < p_1^*$ or $p_2^b < p_2^*$. The reason is that the platform's profit is always non-negative in the bargaining case, while it is negative in the welfare-maximizing case. This also implies that the total welfare is strictly lower under bargaining than in the welfare maximizing benchmark. Therefore, while bargaining can be welfare-improving compared to unilateral price setting, it can not achieve the highest possible total utility.[30]

5 Conclusions

In this paper, I examine the problem of bargaining between one indispensable player and a continuum of fringe players. I model this setting relying on the concept of random order values from cooperative game theory. I demonstrate that random-order-value-based outcomes have some desirable properties,

[28] This is because this term is linear in the number of entrants.

[29] In such a case, the platform would like to set lower entry fees than what it can achieve via bargaining. While in some settings it makes sense to assume that it can do that, in others, commitment issues might prevent it from doing so.

[30] One could argue that the welfare maximizing case is not a realistic benchmark anyway, as it assumes that the platform is making a loss.

and coincide with intuitive expectations in terms of comparative statics. Furthermore, the results are analytically tractable, and have neat interpretations both from a geometric perspective, and in terms of players' marginal contributions.

Random order values offer a comprehensive framework for modeling bargaining outcomes, and also include important and well-known special cases, such as the Shapley value or the weighted value. Due to the former retaining many of the attractive properties of its more specialized counterparts, it might be a good candidate for modeling bargaining outcomes flexibly and in a tractable manner. Furthermore, as shown in Section 2.3, the results can also be generalized to cases with not completely indispensable big players, while retaining the essential structure of the results.

Another contribution of this paper is the derivation of the weighted value in the case of multiple types of fringe players. This provides a rather flexible model (one more general than the Shapley value) for modeling multi-sided markets. The proof of the corresponding theorem is also instructive in that it highlights how bargaining weights affect player's values through determining the probabilities of player orderings.

Finally, I demonstrate how the proposed framework can be applied in practice by examining the two-sided platform model based on Armstrong (2006). I show that the resulting entry fees have some similarities to the welfare-maximizing and unilateral price setting benchmarks, but also differ in some important ways. In particular, the externalities that players entrants impose on the platform (cost of serving) and the other side (network externalities) are not fully incorporated into the entry fees, but shared across the affected players. Similarly, the value generated on each side also factors into the entry fee, contrary to the benchmark cases where it does not directly matter. This suggests, in situations where bargaining between players is plausible, certain effects might be overrepresented or missed if one assumes unilateral price-setting.

There is a variety of ways in which this paper could be expanded or built upon. One obvious direction is to consider other solution concepts from cooperative game theory, such as the nuclelous[31]. Another is to consider either a wider, or different class of games, and see if the various values can be derived in a similar manner. A further direction is to consider is building non-cooperative microfoundations for the proposed random-order-value-based bargaining outcomes. Many such results exist for the Shapley value, and there are also a couple for the the weighted value, but motivating the more general random order value in a similar manner could be a significant step forward for bargaining theory.

Besides these more theoretical directions, the practical applications are no less interesting. The proposed profit sharing framework could be embedded in more applied and detailed models describing various settings. The example application in this paper is a streamlined example of this, while Stancsics (2024) is a further step in this direction, applying these ideas to a model of hybrid platforms. Due to the simplicity of the results, this approach might be a viable, almost drop-in replacement for unilateral price setting in cases where one does not wish to ascribe all the bargaining power to one player.

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[31] It retains many of the desirable properties of the weighted values, such as existence and uniquenes while having a more immediate connection to non-cooperative bargaining theory.

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A Idiosyncratic entry costs are modeled explicitly

Assume that the total utility of an entrant on side i also includes an idiosyncratic cost component. Let us order the players on side i by this idiosyncratic cost in an increasing manner. That is, if we denote the idiosyncratic utility of player n on side i as $\kappa_i(n)$, then κ_i is strictly increasing in n. Also assume that $\lim_{n\to\infty} \kappa_i(n) = \infty$.

Entry is given by the usual assumption that a player enters if and only if the total utility of entering is non-negative. Due to the properties of κ_i, this implies that there is a well-defined, finite number of

entrants on each side. Let us denote the number of entrants as a function of their non-idiosyncratic utility as $\phi_i(u_i)$. Then, the total number of players on side i is

$$n_i = \kappa_i^{-1}(u_i) := \phi_i(u_i).$$

This microfoundation does not change the results of the welfare-maximizing and unilateral pricing cases in any fundamental way. Total welfare must be adjusted to include this idosyncratic cost, but entry fees do not change at all. The reason is that, in the welfare-maximizing case, optimal prices only depend on externalities, while in the unilateral pricing case, the only additional factor is the elasticity of entry[32]. On the other hand, the bargaining case is more interesting. The reason is that this idiosyncratic cost becomes a part of the value that the players bargain over, and thus it affects the characteristic function and the resulting bargaining outcomes.

I approach this problem by breaking it down into two parts: the bargaining over the idisyncratic and non-idiosyncratic parts of the value generated. Due to the linearity property of the Shapley value, the bargaining outcomes over the whole value are simply the sum of the bargaining outcomes over these two parts. Furthermore, the non-idiosyncratic part is simply the value function from the main text, and thus the resulting weighted values are the same as before.

Let us continue by examining the idiosyncratic part of the value. Assume that, even though entrants with different costs have different contribution to the total value, there can be no discrimination between them in terms of the bargaining outcomes. That is, they each must share the same fraction of the total idiosyncratic cost (e.g. have the same value).

Let us denote the total idiosyncratic cost incurred by all players on both sides as $K(n_1, n_2)$. Also, suppose that if n_i players enter on side i, then they are the ones with the lowest idiosyncratic cost. Then, the total idiosyncratic cost is simply the sum of the idiosyncratic costs of the entrants on both sides:

$$K(n_1, n_2) = \int_0^{n_1} \kappa_1(s)\mathrm{d}s + \int_0^{n_2} \kappa_2(s)\mathrm{d}s.$$

One can obtain the weighted values of all sides using Theorem 1.

Proposition A.1. *The weighted values of the idiosyncratic cost component are*

$$K_P(n_1, n_2) = \sum_{i\in\{1,2\}} \int_0^{n_1} \left[1 - \lambda_i \left(\frac{s}{n_i}\right)^{\lambda_P+\lambda_i-1}\right] \kappa_i(s)\mathrm{d}s,$$
$$K_i(n_1, n_2) = \lambda_i \int_0^{n_1} \left(\frac{s}{n_i}\right)^{\lambda_P+\lambda_i-1} \kappa_i(s)\mathrm{d}s.$$

That is, each side only has to bear some part of the total idiosyncratic cost, and the rest is shared with the platform. As with the rest of the value, the share of the idiosyncratic cost is increasing in one's own bargaining weight. This might seem counterintuitive at first, as one may expect that a player with a higher bargaining weight would be able to shift more of the cost to the other side. However, in the case of random order values, this intuition is only correct for monotonic games, which it is not. One has to keep in mind though that this is just part of the final payoffs, and the other parts are increasing in one's own bargaining weight. Therefore, as long as the game over the total value is monotonic, the final payoffs will be increasing in one's own bargaining weight.

Let us finally consider total welfare. As mentioned before, it is simply the sum of the value generated

[32]With this microfoundation in mind, one also relate this entry elasticity as the marginal cost of entry.

and the idiosyncratic cost. That is, the welfare of side $i \in \{1,2\}$ is

$$W_i^b(n_1,n_2) = \frac{\lambda_i}{\lambda_P+\lambda_i+\lambda_j\gamma_i}\alpha_i n_i n_j^{\gamma_i} + \frac{\lambda_i\gamma_j}{\lambda_P+\lambda_j+\lambda_i\gamma_j}\alpha_j n_j n_i^{\gamma_j} - \frac{\lambda_i}{\lambda_P+\lambda_i} n_i F_i$$
$$\underbrace{- \lambda_i \int_0^{n_1} \left(\frac{s}{n_i}\right)^{\lambda_P+\lambda_i-1} \kappa_i(s)\mathrm{d}s}_{(*)}.$$

In turn, the implied entry fee for side i is

$$p_i^b = \frac{\lambda_i}{\lambda_P+\lambda_i}F_i - \frac{\lambda_i\gamma_j}{\lambda_P+\lambda_j+\lambda_i\gamma_j}\alpha_j n_j n_i^{\gamma_j-1}$$
$$+ \frac{\lambda_P+\lambda_j\gamma_i}{\lambda_P+\lambda_i+\lambda_j\gamma_i}\alpha_i n_j^{\gamma_i} - \underbrace{\int_0^{n_1}\left[1-\lambda_i\left(\frac{s}{n_i}\right)^{\lambda_P+\lambda_i-1}\right]\kappa_i(s)\mathrm{d}s}_{(**)}.$$

As with the other components of the value, the idiosyncratic entry cost is also shared between the platform and the entrants. Part of it $(*)$ is incurred on the side of the entrants, while the other part $(**)$ paid by the platform as a reduction of entry fees. How this cost is divided is not straightforward, though. As in the case of non-linear network effects, it depends on the bargaining weights and the shape of the idiosyncratic cost function, with the two also interacting in a non-trivial way.

B Miscellaneous lemmas

Lemma B.1. *Let*

$$\Delta_n(s) = \frac{\log(s+1/n)-\log(\lambda)}{n},$$
$$\Delta(s) = \frac{1}{s}.$$

Then $\Delta_n \xrightarrow{\mathrm{u}} \Delta$ *uniformly on* $[t,1]$ *for any* $t>0, \lambda>0$.

Proof of Lemma B.1. First, note that Δ_n and Δ are all continuous functions. Then, following the standard proof for $\frac{\mathrm{d}}{\mathrm{d}s}log(s) = \frac{1}{s}$ rewrite $\Delta_n(s)$ as

$$\Delta_n(s) = \frac{\log(s+1/n)-\log(\lambda)}{n}$$
$$= \log\left(1+\frac{1}{sn}\right)^n.$$

It is well known that $\left(1+\frac{1}{sn}\right)^n$ is monotone increasing in n and converges to $\exp(1/s)$. Therefore, the pointwise convergence of $\Delta_n(s) \to \Delta(s)$ is also monotone.

Finally, by Dini's theorem, the monotone pointwise convergence of a sequence of continuous functions to a continuous function on a compact set implies uniform convergence on that set. □

Lemma B.2. *Let* $f_n, f : [a,b] -> \mathbb{R}$ *be Riemann-integrable functions with* $f_n \xrightarrow{\mathrm{u}} f$ *uniformly. Then,*

$$\lim_{n\to\infty} \frac{b-a}{n}\sum_{k=1}^{n} f_n\left(a+\frac{b-a}{n}\right) = \int_0^1 f(t)\mathrm{d}t.$$

Proof of Lemma B.2.

$$
\begin{aligned}
&\lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} f_n\left(a + \frac{b-a}{n}\right) \\
&= \lim_{n\to\infty} \frac{b-a}{n} \left[\sum_{k=1}^{n} f\left(a + \frac{b-a}{n}\right) + \sum_{k=1}^{n} \left(f_n\left(a + \frac{b-a}{n}\right) - f\left(a + \frac{b-a}{n}\right)\right)\right] \\
&= \int_a^b f(t)\mathrm{d}t + \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} \left(f_n\left(a + \frac{b-a}{n}\right) - f\left(a + \frac{b-a}{n}\right)\right) \\
&\leq \int_a^b f(t)\mathrm{d}t + \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} \left|f_n\left(a + \frac{b-a}{n}\right) - f\left(a + \frac{b-a}{n}\right)\right| \\
&\leq \int_a^b f(t)\mathrm{d}t + \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} \sup_{t\in[a,b]} |f_n(t) - f(t)| \\
&= \int_a^b f(t)\mathrm{d}t + (b-a) \underbrace{\lim_{n\to\infty} \sup_{t\in[a,b]} |f_n(t) - f(t)|}_{=0 \text{ due to uniform convergence}} \\
&= \int_a^b f(t)\mathrm{d}t
\end{aligned}
$$

□

Lemma B.3. *Consider the random vector X_n with values from $[0,1]^L$. Let S, be a random variable with support $[0,1]$ and cumulative distribution function $G_n(s)$. Assume that for every $s \in [0,1]$, $X_n \mid S = s \xrightarrow{a.s.} h(s)$ where $h(s)$ is some continuous function. Then $X_n \xrightarrow{d} h(S)$.*

Proof of Lemma B.3. We have to show that

$$\lim_{n\to\infty} \Pr(X_n \leq x) = \Pr(h(S) \leq x).$$

First, note that

Let us start by conditioning on S:

$$\Pr(X_n \leq x) = \int_0^1 \Pr(X_n \leq x \mid S = s)\mathrm{d}G(s).$$

Now consider the following:

$$
\begin{aligned}
\lim_{n\to\infty} |\Pr(X_n \leq x) - \Pr(h(S) \leq x)| &= \lim_{n\to\infty} \left|\int_0^1 \Pr(X_n \leq x \mid S = s)\mathrm{d}G(s) - \Pr(h(S) \leq x)\right| \\
&= \lim_{n\to\infty} \left|\int_0^1 \Pr(X_n \leq x \mid S = s)\mathrm{d}G(s) - \int_0^1 \mathbf{1}[h(s) \leq 1]\mathrm{d}G(s)\right| \\
&= \lim_{n\to\infty} \left|\int_0^1 \Pr(X_n \leq x \mid S = s) - \mathbf{1}[h(s) \leq x]\mathrm{d}G(s)\right| \\
&\leq \lim_{n\to\infty} \int_0^1 |\Pr(X_n \leq x \mid S = s) - \mathbf{1}[h(s) \leq x]|\,\mathrm{d}G(s) \\
&= \int_0^1 \lim_{n\to\infty} |\Pr(X_n \leq x \mid S = s) - \mathbf{1}[h(s) \leq x]|\,\mathrm{d}G(s) \\
&= 0
\end{aligned}
$$

for all $x \in [0,1]^L$. The exchange of the limit and the integral is justified by the fact that the integrand is dominated by a constant function, and therefore the dominated convergence theorem applies. The last equality follows from the fact that $X_n \mid S = s \xrightarrow{a.s.} h(s)$. □

C Proofs of propositions in the main text

Proof of Proposition 1. Monotonicity is evident. For superadditivity, note that for any coalitions S_1, s_2 such that $S_1 \cap s_2 = \emptyset$, $P \notin S_1$ or $P \notin s_2$. WLOG assume it is the latter, therefore $v(S_2) = 0$. As a result, $v(S_1) + v(S_2) = v(S_1) \leq v(S_1 \cup S_2)$ holds if and only if (N, v) is monotone. □

Proof of Proposition 2. First, observe that f is continuous on the compact set $[0,1]$, and is therefore also bounded. Furthermore,

$$\varphi_P^n = \frac{1}{n+1} \sum_{k=0}^{n} \Pr(|\mathcal{P}_P|/n = k) f(k/n) = \mathbb{E}[X_n].$$

As f is continuous and bounded, and $X_n \xrightarrow{d} X$, by the portmanteau lemma, $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$. Putting it together,

$$\begin{aligned}
\lim_{n\to\infty} \varphi_P^n &= \frac{1}{n+1} \sum_{k=0}^{n} \Pr(|\mathcal{P}_P|/n = k) f(k/n) \\
&= \lim_{n\to\infty} \mathbb{E}[X_n] \\
&= \mathbb{E}[X] \\
&= \int_0^1 f(t) \mathrm{d}G(t).
\end{aligned}$$

Furthermore, if G is differentiable, then

$$\int_0^1 f(t) \mathrm{d}G(t) = \int_0^1 g(t) f(t) \mathrm{d}t.$$

□

Proof of Proposition 3. Let R denote a permutation of the set of players (N). Additionally, let us denote the players preceding i by $\mathcal{P}_i^R$. The value of player P is their expected marginal contribution averaged over all permutations of N:

$$\varphi_P^n = \frac{1}{(n+1)!} \sum_R v(\mathcal{P}_P^R \cup \{i\}) - v(\mathcal{P}_P^R)$$

First, note that $v(\mathcal{P}_P^R) = 0$ for any permutation, as no coalition can achieve a positive value without player P. Furthermore, using the fact that all agents of type A are identical implies that $v(\mathcal{P}_P^R \cup \{i\})$ only depends on the number of agents in the coalition. More precisely,

$$v(\mathcal{P}_P^R \cup \{i\}) = f(n_A(\mathcal{P}_P^R \cup \{i\})/n) = f(|\mathcal{P}_P^R|/n).$$

Finally, the set of permutations in which k number of players precede P is independent of n, i.e.

$$\{R \mid |\mathcal{P}_P^R| = k\} = n! \quad \forall k.$$

Putting all the above together, the value of player P can be expressed as

$$\begin{aligned}
\varphi_P^n &= \frac{1}{(n+1)!}\sum_{k=0}^{n} n! f(k/n) \\
&= \frac{1}{n+1}\sum_{k=0}^{n} f(k/n) \\
&= \frac{n}{n+1}\underbrace{\frac{1}{n}\sum_{k=0}^{n-1} f(k/n)}_{=S_n} + \frac{1}{n+1} f(1).
\end{aligned}$$

S_n are just the left Riemann-sums of function f on the interval $[0,1]$. Therefore, if f is continuous (and thus Riemann-integrable), then $S_n \to \int_0^1 f(t)$, and thus

$$\begin{aligned}
\lim_{n\to\infty} \varphi_P^n &= \lim_{n\to\infty} \frac{1}{n+1}\sum_{k=1}^{n} f(k/n) \\
&= \lim_{n\to\infty} \underbrace{\frac{n}{n+1}}_{\to 1} \frac{1}{n}\sum_{k=0}^{n-1} f(k/n) + \underbrace{\frac{1}{n+1} f(1)}_{\to 0} \\
&= \int_0^1 f(t)\mathrm{d}t.
\end{aligned}$$

□

Proof of Corollary 2. The first equality comes from the efficiency of the Shapley-value. The values of all players sum up to $f(1)$ for all $n \in \mathbb{N}$, therefore

$$\lim_{n\to\infty} \sum_{i=1}^{n} \varphi_{A_i}^n = \lim_{n\to\infty} (1 - \varphi_P^n) = 1 - \int_0^1 f(t).$$

The second one can be obtained by integration by parts:

$$\int_0^1 t f'(t)\mathrm{d}t = t f(t) \mid_0^1 - \int_0^1 f(t)\mathrm{d}t = f(1) - \int_0^1 f(t)\mathrm{d}t$$

□

Proof of Lemma 1. The probability of P having at most fraction t of the other players before itself is

$$\begin{aligned}
\Pr(X_n \le nt) &= \Pr(X_n \le nt) \\
&= \prod_{j=nt+1}^{n} \frac{j}{j+\lambda} \\
&= \exp\Bigg(\underbrace{\sum_{j=nt+1}^{n} \log(j) - \log(j+\lambda)}_{\equiv S_n}\Bigg).
\end{aligned}$$

Taking limits,

$$\begin{aligned}
\lim_{n\to\infty} S_n &= \lim_{n\to\infty} \sum_{j=nt+1}^{n} \log(j) - \log(j+\lambda) \\
&= \lim_{n\to\infty} \sum_{i=1}^{n-nt} \log(nt+i) - \log(nt+i+\lambda) \\
&= \lim_{n\to\infty} \frac{1}{n-nt} \sum_{i=1}^{n-nt} \frac{\log\left(t+\frac{i}{n-nt}\right) - \log\left(t+\frac{i}{n-nt}+\frac{\lambda}{n-nt}\right)}{1/(n-nt)}
\end{aligned}$$

Let

$$\Delta_n(s) = \frac{\log(s) - \log\left(s+\frac{\lambda}{n-nt}\right)}{1/(n-nt)}$$

By lemma B.1,

$$\Delta_n \xrightarrow{\text{u}} \lambda \frac{\mathrm{d}}{\mathrm{d}s} log(s) = -\frac{\lambda}{s}$$

on the compact interval $[t, 1]$ for any $t > 0$ ($\xrightarrow{\text{u}}$ denotes uniform convergence).

Then, by lemma B.2, we have that

$$\begin{aligned}
\lim_{n\to\infty} S_n &= \lim_{n\to\infty} \frac{1}{n-nt} \sum_{i=1}^{n-nt} \Delta_n\left(t+\frac{i}{n-nt}\right) \\
&= \int_t^1 (\lim_{n\to\infty} \Delta_n)(s)\mathrm{d}s \\
&= \int_t^1 \lim_{n\to\infty} -\frac{\lambda}{s}\mathrm{d}s \\
&= \lambda \log t
\end{aligned}$$

Substituting $\lim_{n\to\infty} S_n$ into the original equation yields

$$\begin{aligned}
\lim \Pr\left(\frac{X_n}{n} \le t\right) &= \exp\left(\lim_{n\to\infty} \sum_{j=nt+1}^{n} \log(j) - \log(j+\lambda)\right) \\
&= \exp(\lambda \log(t)) \\
&= t^\lambda.
\end{aligned}$$

For $t = 0$, simply observe that

$$\lim_{n\to\infty} \prod_{j=1}^{n} \frac{j}{j+\lambda} = 0 = 0^\lambda.$$

□

Proof of Corollary 3. If f is monotone increasing, then $f(0) \le \int_0^1 f(t)\mathrm{d}t \le f(1)$. The inequalities become strict if f is not constant on the whole interval. □

Proof of Proposition 4. The weighted value of player P is its expected contribution across all permutations,

with each permutation weighted by its probability of occurring.

$$\varphi_P^{n,\lambda} = \sum_R \Pr(R)[v(\mathcal{P}_P^R \cup \{i\}) - v(\mathcal{P}_P^R)]$$

As before, using the fact that fringe players are identical, this can be rephrased as

$$\begin{aligned}\varphi_P^{n,\lambda} &= \sum_{k=0}^{n} \Pr(R) f(k/n) \\ &= \mathbb{E}[f(X_n/n)]\end{aligned}$$

where X_n is defined as above. f is continuous, and therefore bounded on the compact set $[0,1]$. As a consequence, $\frac{X_n}{n} \xrightarrow{d} X$ implies $\mathbb{E}[f(X_n/n)] \to \mathbb{E}[f(X)]$, which in turn gives

$$\begin{aligned}\lim_{n\to\infty} \varphi_P^{n,\lambda} &= \lim_{n\to\infty} \mathbb{E}[f(X_n/n)] \\ &= \mathbb{E}[f(X)] \\ &= \int_0^1 f(t)\mathrm{d}G(t) \\ &= \int_0^1 g(t)f(t)\mathrm{d}t\end{aligned}$$

where $G(t)$ and $g(t)$ are the cdf and pdf of X, respectively. □

Proof of Corollary 4. Let X and X' be random variables with cdfs $G_X = t^\lambda$ and $G_{X'} t^{\lambda'}$, respectively. For any $\lambda < \lambda'$, $t^\lambda > t^{\lambda'} \forall t \in [0,1]$, meaning that X' first-order stochastically dominates X'. As a result, for any monotonically increasing f,

$$\int_0^1 g_X(t)f(t)\mathrm{d}t = \mathbb{E}[f(X)] \leq \mathbb{E}[f(X')] = \int_0^1 g_{X'}(t)f(t)$$

with strict inequality unless f is constant almost everywhere. As f is continuous, the latter is equivalent to f being constant on the whole $[0,1]$ interval. □

Proof of Corollary 5. As $\lambda \to 0$, X converges to the degenerate random variable X_0 for which $\Pr(X_0 = 0) = 1$. As a consequence, the expected value of $f(X)$ converges to $\mathbb{E}[f(X_0)] = f(0)$. $\lim_{\lambda\to\infty} \varphi_P^{\infty,\lambda} = f(1)$ can be shown along the same lines. □

Proof of **??**. First, let us rewrite the expected marginal contribution of player P_i in the following way:

$$\begin{aligned}\varphi_{P_i}^{n,m} &= \frac{1}{(n+m)!} \sum_R v(\mathcal{P}_{P_i}^R \cup \{i\}) - v(\mathcal{P}_{P_i}^R) \\ &= \frac{1}{(n+m)!} \left[\sum_{\{R | n_P(\mathcal{P}_{P_i}^R = 0)\}} \left[v(\mathcal{P}_{P_i}^R \cup \{i\}) - v(\mathcal{P}_{P_i}^R)\right] + \sum_{\{R | n_P(\mathcal{P}_{P_i}^R > 0)\}} \left[v(\mathcal{P}_{P_i}^R \cup \{i\}) - v(\mathcal{P}_{P_i}^R)\right] \right]\end{aligned}$$

Next, make use of the fact that the marginal contribution of a platform is zero if another platform is

already present, and only depends on the number of fringe firms otherwise.

$$\begin{aligned}\varphi_{P_i}^{n,m} &= \frac{1}{(n+m)!}\sum_{\{R|n_P(\mathcal{P}_{P_i}^R=0)\}} f(\mathcal{P}_A^R/n)\\ &= \frac{1}{(n+m)!}\sum_{k=0}^{n}|\{R|n_A(\mathcal{P}_{P_i}^R=k) \text{ and } n_P(\mathcal{P}_{P_i}^R=0)\}|f(k/n)\end{aligned}$$

Finally, notice that, while the permutations with k fringe agents before P_i does not depend on k, amongst those, the number of permutations with no platform before P_i does depend on it. In particular,

$$|\{R|n_A(\mathcal{P}_{P_i}^R=k) \text{ and } n_P(\mathcal{P}_{P_i}^R=0)\}| = \frac{n!(n+m-k-1)!}{(n-k)!}$$

and

$$\begin{aligned}\varphi_{P_i}^{n,m} &= \sum_{k=0}^{n}\frac{n!(n+m-k-1)!}{(n+m)!(n-k)!}f(k/n)\\ &= \sum_{k=0}^{n}\frac{(n-k+1)(n-k+2)\dots(n-k+m-1)}{(n+1)(n+2)\dots(n+m)}f(k/n)\\ &= \frac{1}{n+1}\sum_{k=0}^{n}\underbrace{\frac{(n(1-k/n)+1)(n(1-k/n)+2)\dots(n(1-k/n)+m-1)}{(n+2)\dots(n+m)}f(k/n)}_{S_n}\end{aligned}$$

For any $t=k/n\in[0,1]$,

$$S_n(t)\to(1-t)^{m-1}f(t)\equiv S(t).$$

Furthermore, it is simple to verify that this convergence is uniform, and that S_n and S are all continuous functions. By Dini's theorem, the former properties imply that $S_n \xrightarrow{\text{u}} S$ uniformly on $[0,1]$.

The conclusion of the proof is similar to that of proposition 3:

$$\begin{aligned}\lim_{n\to\infty}\varphi_{P_i}^{n,m} &= \lim_{n\to\infty}\underbrace{\frac{n}{n+1}}_{\to 1}\sum_{k=0}^{n-1}S_k(k/n)+\underbrace{\frac{1}{1+n}S(1)}_{\to 0}\\ &= \lim_{n\to\infty}\sum_{k=0}^{n-1}S_k(k/n)\\ &= \int_0^1(1-t)^{m-1}f(t)\mathrm{d}t\end{aligned}$$

with the last equality supported by lemma B.2. □

Proof of Corollary 10. The allocation is efficient for all $n\in\mathbb{N}$, therefore efficient in the limit, as well. The second equality can be obtained by integration by parts. □

Proof of Corollary 9. (Assuming f is continuously differentiable) $f'(t)$ is non-negative, so $[1-(1-t)^m]f'(t)$ is increasing in m $\forall t\in[0,1]$. As a consequence, $\varphi_A^{n,\infty}$ is also increasing in m if $f'(t)$ is positive on some interval or, $f(t)$ is not constant. Conversely, $\varphi_P^{\infty,m}=1-\varphi_A^{\infty,m}$ is decreasing in m. □

Proof of Proposition 10. The proof mirrors that of Proposition 2. First, note that the random order value

can be expressed as follows:

$$\begin{aligned}
\varphi_P^n &= \sum_{k_1=0}^{n} \cdots \sum_{k_L=0}^{n} \Pr(n_{A_1}(\mathcal{P}_P) = k_1, \ldots, n_{A_L}(\mathcal{P}_P) = k_L) f\left(\frac{k_1}{n}, \ldots, \frac{k_L}{n}\right) \\
&= \sum_{k_1=0}^{n} \cdots \sum_{k_L=0}^{n} \Pr(nX_n = (k_1, \ldots, k_L)) f\left(\frac{k_1}{n}, \ldots, \frac{k_L}{n}\right) \\
&= \sum_{k_1=0}^{n} \cdots \sum_{k_L=0}^{n} \Pr\left(X_n = \left(\frac{k_1}{n}, \ldots, \frac{k_L}{n}\right)\right) f\left(\frac{k_1}{n}, \ldots, \frac{k_L}{n}\right) \\
&= \mathbb{E}[f(X_n)].
\end{aligned}$$

Furthermore, as f is continuous (and therefore bounded on $[0,1]^L$), and $X_n \xrightarrow{d} X$, by the portmanteau lemma, $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$. Putting it together,

$$\begin{aligned}
\lim_{n\to\infty} \varphi_P^n &= \lim_{n\to\infty} \mathbb{E}[f(X_n)] \\
&= \mathbb{E}[f(X)] \\
&= \int_0^1 \cdots \int_0^1 f(t_1, \ldots, t_L) \mathrm{d}G(t_1, \ldots t_L).
\end{aligned}$$

Finally, if X is a continuous random variable, then

$$\int_0^1 \cdots \int_0^1 f(t_1, \ldots, t_L) \mathrm{d}G(t_1, \ldots t_L) = \int_0^1 \cdots \int_0^1 g(t_1, \ldots, t_L) f(t_1, \ldots, t_L) \mathrm{d}t_1 \ldots \mathrm{d}t_L.$$

□

Proof of Lemma 2. Just like in the proof of Proposition 10, the random order value can be expressed as follows:

$$\varphi_P^n = \mathbb{E}[f(X_n)].$$

Furthermore, as f is continuous and bounded on $[0,1]^L$, and $X_n \xrightarrow{d} X$, by the portmanteau lemma, $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$. Therefore,

$$\begin{aligned}
\lim_{n\to\infty} \varphi_P^n &= \lim_{n\to\infty} \mathbb{E}[f(X_n)] \\
&= \mathbb{E}[f(X)] \\
&= \mathbb{E}[f(a_1(\xi), \ldots, a_L(\xi))] \\
&= \int_0^1 f(a_1(s), \ldots a_L(s)) \mathrm{d}H(s).
\end{aligned}$$

For the fringe value, the proof is similar, with a couple of additional steps. First, fix some fringe player A_{li}. Let Y_n be the random variable describing the share of each type of fringe player that precede player A_{li}. Also, let Q_n be the random variable describing whether player P precedes player A_{li}. Then, the value of fringe player A_{li} is

$$\varphi_{A_{li}}^n = \mathbb{E}[\mathbf{1}[Q_n = 1](f(Y_1, \ldots, Y_l + 1/n, \ldots, Y_L) - f(Y_1, \ldots, Y_l, \ldots, Y_L))],$$

and the value of the whole fringe of type A_l is

$$\varphi^n_{A_l} = n\varphi^n_{A_{li}} = \mathbb{E}\Bigg[\underbrace{\mathbf{1}[Q_n = 1]\frac{f(Y_1,\dots,Y_l+1/n,\dots,Y_L) - f(Y_1,\dots,Y_l,\dots,Y_L)}{1/n}}_{:=\Delta_n}\Bigg].$$

Now notice that, because the number of fringe players goes to infinity,

$$\begin{aligned} X_n \xrightarrow{d} (a_1(\xi),\dots,a_L(\xi)) &\implies Y_n \xrightarrow{d} (a_1(\xi),\dots,a_L(\xi)) \\ &\implies (Y_1,\dots,Y_n)|Y_l = a_l(s) \xrightarrow{d} (a_1(s),\dots,a_l(s)). \end{aligned}$$

Furthermore,

$$\Pr(Q_n = 1|[Y_n]_l = a_l(s)) \to H(s).$$

Finally, for fixed $(Y_1,\dots,Y_L)$,

$$\Delta_n \to \partial_l f(Y_1,\dots,Y_l,\dots,Y_L).$$

Putting it all together,

$$\begin{aligned} \lim_{n\to\infty} \varphi^n_{A_l} &= \lim_{n\to\infty} \mathbb{E}[\Delta_n(Y_n)] \\ &= \mathbb{E}[H(s)\partial_l f(Y_1,\dots,Y_L)] \\ &= \int_0^1 H(s)\partial_l f(a_1(s),\dots,a_L(s))\mathrm{d}a_l(s) \\ &= \int_0^1 H(s)a_l'(s)\partial_l f(a_1(s),\dots,a_L(s))\mathrm{d}s. \end{aligned}$$

□

Proof of Proposition 11. Let us show than X_n converges in distribution to the degenerate random variable $X = (U,\dots,U)$, where U is uniformly distributed on $[0,1]$.

First, let us switch to an alternative formulation of the problem. Remember, that the probability of each permutation is the same. Therefore, the following procedure leads to the same distribution over permutations: (1) for each player, draw a random number from the uniform distribution on $[0,1]$, then sort the players according to the drawn numbers. This is equivalent to the original problem, as the probability of each permutation is the same.

Now let us characterize X_n in terms of this procedure. $[X_n]_l$ is the proportion of players of type A_l that are placed before player P, or, equivalently, the number of players of type A_l that have drawn a number less than player P. I.e.,

$$[X_n]_l = \frac{1}{n}\sum_{i=1}^n \mathbf{1}[U_{li} < U_P].$$

For any finite n, we can disregard ties, as they have probability zero.

Now let us condition this probability distribution on the draw of player P:

$$[X_n]_l < x \mid U_P = u = \frac{1}{n}\sum_{i=1}^n \mathbf{1}[U_{li} < u].$$

As the draws are independent, by the strong law of large numbers,

$$[X_n]_l < x \mid U_P \xrightarrow{a.s.} \mathbb{E}[\mathbf{1}[U_{li} < u]] = u.$$

Now we can use Lemma B.3 to conclude that $X_n \xrightarrow{d} (U, \ldots, U)$. Finally, by Lemma 2, the Shapley-value is

$$\varphi_P^\infty = \int_0^1 f(s, \ldots, s)\mathrm{d}t$$

and the value of fringe l is

$$\varphi_{A_l}^\infty = \int_0^1 \partial_l f(s, \ldots, s)\mathrm{d}s$$

□

Proof of Theorem 1. The proof uses the same approach as that of Proposition 11. I will show that $X_n \xrightarrow{d} h(S)$, where $h(s) = (s^{\lambda_1}, \ldots, s^{\lambda_L})$, and S is a random variable with cdf $G(s) = s^{\lambda_P}$, and then rely on Lemma 2.

First, as before, let us characterize the probabilities of the various permutations. The approach I am using relies on two main ideas.

First, as shown by Kalai and Samet (1987), the probability of a permutation can be described by successively selecting the players with probabilities proportional to the weights of the players that have not been selected yet, and then reversing this ordering.[33]

The other key result I rely on is the main proposition of Efraimidis and Spirakis (2006). It states that weighted random sampling without replacement can be also be achieved by the following procedure. First, for each player i, draw a random number from the distribution $U^{\frac{1}{\lambda_i}}$, where U is uniformly distributed on $[0, 1]$, and λ_i is the sampling weight of player i. Then, for a sample size of m, select the m players with the largest numbers.

An immediate corollary of the latter result is that the procedure described by Kalai and Samet (1987) is equivalent to the following procedure. First, for each player i, draw a random number from the distribution $V_i = U^{\frac{1}{\lambda_i}}$, where U is uniformly distributed on $[0, 1]$. Then, for each player i, sort the players according to the drawn numbers in ascending order.

Now let us characterize X_n in terms of this procedure. $[X_n]_l$ is the proportion of players of type A_l that are placed before player P, or, equivalently, the number of players of type A_l that have drawn a number less than player P. I.e.,

$$[X_n]_l = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[V_{li} < V_P].$$

Now let us condition this probability distribution on the draw of player P:

$$[X_n]_l < x \mid V_P = s = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[V_{li} < u].$$

Then use the strong law of large numbers to deduce that

$$[X_n]_l < x \mid V_P \xrightarrow{a.s.} \mathbb{E}[\mathbf{1}[V_{li} < u]] = u^{\lambda_l}.$$

[33] For more details, see Section 2.2.2.

Then we can rely on Lemma B.3 to conclude that $X_n \xrightarrow{d} (V_P^{\lambda_1}, \dots, V_P^{\lambda_L})$.

Finally, by Lemma 2, the weighted value of player P is

$$\begin{aligned}\varphi_P^\infty &= \int_0^1 f(s_1^{\lambda_1}, \dots, s_L^{\lambda_L}) \mathrm{d}V_P(s) \\ &= \int_0^1 f(s_1^{\lambda_1}, \dots, s_L^{\lambda_L}) \lambda_P s^{\lambda_P - 1} \mathrm{d}s,\end{aligned}$$

and the value of fringe l is

$$\varphi_{A_l}^\infty = \int_0^1 t^{\lambda_P} \lambda_l s^{\lambda_l - 1} \partial_l f(s_1^{\lambda_1}, \dots, s_L^{\lambda_L}) \lambda_P s^{\lambda_P - 1} \mathrm{d}s.$$

□

Proof of Proposition 12. The (marginal) cost of serving an entrant on side i is F_i. The marginal network externality it generates on the other side is

$$n_j \frac{\partial u_j}{\partial n_i} = \alpha_j \gamma_j n_j n_i^{\gamma_j - 1}.$$

Welfare is maximized when the cost of entry equals the externality from entry, i.e.:

$$p_i^* = F_i - \alpha_j \gamma_j n_j n_i^{\gamma_j - 1}$$

□

Proof of Proposition 13. By assumption, $\pi_P(u_1, u_2)$ is concave in u_1 and u_2, therefore the first-order conditions are necessary and sufficient for a maximum. Simply taking the partial derivatives with respect to u_i yields the following condition for the optimal entry fees:

$$p_i^u = F_i - \alpha_j \gamma_j n_j n_i^{\gamma_j - 1} + \frac{\phi_1(u_1)}{\phi_1'(u_1)}.$$

□

Proof of Proposition 14. From Theorem 1, the value of player P is

$$\begin{aligned}\pi_P &= \int_0^1 \lambda_P t^{\lambda_P - 1} w(t^{\lambda_1} n_1, t^{\lambda_2} n_2) \mathrm{d}t \\ &= \int_0^1 \lambda_P t^{\lambda_P - 1} \left[\alpha_1 t^{\lambda_1} n_1 t^{\lambda_2 \gamma_1} n_2^{\gamma_1} + \alpha_2 t^{\lambda_2} n_2 t^{\lambda_1 \gamma_2} n_1^{\gamma_2} + t^{\lambda_1} F_1 n_1 + t^{\lambda_2} F_2 n_2\right] \mathrm{d}t \\ &= \frac{\lambda_P}{\lambda_P + \lambda_1 + \lambda_2 \gamma_1} \alpha_1 n_1 n_2^{\gamma_1} + \frac{\lambda_P}{\lambda_P + \lambda_2 + \lambda_1 \gamma_2} \alpha_2 n_2 n_1^{\gamma_2} - \frac{\lambda_P}{\lambda_P + \lambda_1} n_1 F_1 - \frac{\lambda_P}{\lambda_P + \lambda_2} n_2 F_2.\end{aligned}$$

Similarly, the value of side $i \in \{1, 2\}$ is

$$\begin{aligned}W_i &= \int_0^1 t^{\lambda_P} \lambda_i t^{\lambda_i - 1} \partial_i w(t^{\lambda_1} n_1, t^{\lambda_2} n_2) \mathrm{d}t \\ &= \int_0^1 t^{\lambda_P} \lambda_i t^{\lambda_i - 1} \left[\alpha_i n_i t^{\lambda_j \gamma_i} n_j^{\gamma_i} + \alpha_j \lambda_i \gamma_j t^{\lambda_j} n_j t^{\lambda_i (\gamma_j - 1)} n_i^{\gamma_j} - n_i F_i\right] \mathrm{d}t \\ &= \frac{\lambda_i}{\lambda_P + \lambda_i + \lambda_j \gamma_i} \alpha_i n_i n_j^{\gamma_i} + \frac{\lambda_i \gamma_j}{\lambda_P + \lambda_j + \lambda_i \gamma_j} \alpha_j n_j n_i^{\gamma_j} - \frac{\lambda_i}{\lambda_P + \lambda_1} n_i F_i.\end{aligned}$$

□

Proof of Theorem 2. See the proof of Propositions 12 to 14. □